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PAPER

Observer-robust energy condition verification for warp drive spacetimes

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Abstract
Energy-condition tests for warp-drive spacetimes must account for all admissible causal directions at each point. We use the classical S-lemma to express the null, weak, and strong conditions as 4×4 linear matrix inequalities; the dominant condition requires two such tests. These criteria require neither Hawking–Ellis classification nor a rapidity cutoff. For the null condition in a fixed orthonormal tetrad, the optimized multiplier margin equals half the minimum normalized null-energy contraction. Interval evaluation of the metric and curvature provides pointwise certificates and global bounds on its minimum over the bubble wall. On flat unit-lapse slices, the momentum constraint relates Eulerian momentum to shift vorticity. For smooth shifts on all of Euclidean space with bounded vorticity, momentum vanishes identically precisely for a gradient plus rigid rotation. For shifts linear in speed, integrated negative Eulerian energy scales exactly quadratically when finite, with a fixed profile and integration domain on these slices. We implement the tests in WARPAX, a JAX toolkit, and compare four warp-drive geometries at matched parameters across subluminal and superluminal speeds. At the reference parameters, the global bounds establish null-energy violation in all four bubble walls. The ideal irrotational Rodal profile has zero Eulerian momentum and is everywhere Type I; Type-IV regions are detected in the other sampled walls. For Rodal, the Eulerian reading misses about 73% of the sampled wall weak-energy violations. These numerical type labels and sampled fractions are distinct from the interval bounds. Finite-segment null-geodesic integrals and flat-space quantum-inequality estimates supplement the pointwise analysis.

Keywords: warp drive, energy conditions, Hawking–Ellis classification, general relativity, automatic differentiation, JAX

1 Introduction

Alcubierre's 1994 warp drive metric [1] demonstrated that general relativity admits solutions describing effective superluminal travel by deforming the spacetime geometry around a compact region, a "warp bubble". The bubble center follows a timelike worldline while the bubble moves at an arbitrary coordinate velocity v_s through the asymptotically Minkowski exterior. The metric takes the Arnowitt–Deser–Misner (ADM) $3+1$ form

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + (\mathrm{d}x - v_s f(r_s)\,\mathrm{d}t)^2 + \mathrm{d}y^2 + \mathrm{d}z^2\,, \tag{1}$$

where $v_s = \dot{x}_s(t)$ and $f(r_s)$ is a smooth top-hat profile, approximately unity inside the bubble and tending to zero outside. Here $r_s = [(x - x_s(t))^2 + y^2 + z^2]^{1/2}$ is the distance from the bubble center.

A central obstacle for the warp metrics considered here is their requirement for exotic matter: matter whose stress-energy tensor violates the classical energy conditions. Throughout this paper, the spacetime signature is $(-+++)$ and we work in geometric units ($G = c = 1$). The four energy conditions, namely the null (NEC), weak (WEC), strong (SEC), and dominant (DEC), are pointwise inequalities on the stress-energy tensor contracted with timelike or null vectors:

$$\text{NEC:}\quad T_{ab}\,k^a k^b \geq 0 \qquad \forall \text{ null } k^a\,, \tag{2}$$

$$\text{WEC:}\quad T_{ab}\,u^a u^b \geq 0 \qquad \forall \text{ timelike } u^a\,, \tag{3}$$

$$\text{SEC:}\quad \left(T_{ab} - \tfrac{1}{2}\,T\,g_{ab}\right) u^a u^b \geq 0 \qquad \forall \text{ timelike } u^a\,, \tag{4}$$

$$\text{DEC:}\quad T_{ab}\,u^a u^b \geq 0 \text{ and } -T^a{}_b\,u^b \text{ future causal} \qquad \forall \text{ future timelike } u^a\,. \tag{5}$$

Here $T = g^{ab}T_{ab}$ is the trace, and the DEC permits zero energy flux. These conditions must hold for *all* admissible observers at each spacetime point [2, 3]. An energy condition either holds or fails at a point independently of the observer. A single-frame *diagnostic*, such as $T_{ab}u^a u^b$ for a particular u^a, depends on that observer.

For a Type I stress-energy tensor, the energy conditions reduce to inequalities on the invariant rest-frame energy density ρ and principal pressures p_i; the eigenvalues of $T^a{}_b$ are $-\rho$ and the p_i. A Type IV tensor has no causal eigenvector and violates the NEC [2, 3, 5, 14]. Section 2.2 states these classical criteria and derives the momentum discriminant under its principal-axis hypothesis. Theorem 1 gives a complementary formulation: 4×4 linear matrix inequalities decide all four conditions at every algebraic type, including Types II and III. These criteria require neither a rapidity cap nor a timelike coordinate vector ∂_t. Corollary 2 relates the optimized multiplier to the null-cone minimum in a fixed tetrad. Interval evaluation of the metric and its derivatives then gives bounds on that minimum. The pointwise interval tests decide all four conditions at each of 4257 sampled wall points per drive (Appendix E.2); the separate global search encloses the wall infimum of the normalized null contraction (Appendix C).

The algebraic type is related to the Eulerian momentum density j. When its direction is a principal axis of the spatial stress, the sign of $\Delta = (\rho_n + S_\parallel)^2 - 4|j|^2$ determines the Type-I/II/IV transition. Transverse stress coupling requires the separate reduction of Corollary 1. On time-independent flat unit-lapse slices, $8\pi|j| = \frac{1}{2}|\nabla \times (\nabla \times \beta)|$. For a smooth shift on $\mathbb{R}^3$ with bounded vorticity, vanishing momentum characterizes gradient shifts up to rigid rotation (Lemma 2). This result concerns momentum; the observed vorticity/type association is empirical. For speed-linear shifts, the same constraints give an exact quadratic law for the integrated negative Eulerian energy on a fixed domain (Lemma 4). A quadratic law for the pointwise eigenvalue margin follows under the additional assumption $j = 0$.

We compare four drives at matched parameters: Alcubierre, Natário, Van den Broeck, and Rodal, using the laboratory-frame conventions of Appendix D. The ideal irrotational Rodal geometry is Type I everywhere at every speed (Lemma 3). The sampled Alcubierre and Natário walls contain Type-IV regions; Van den Broeck's wall Type-IV fraction first crosses 50% at $v_s \approx 0.38$. Within Rodal's wall the single-frame Eulerian diagnostic does not register ≈74% of the dominant and ≈73% of the weak energy-condition violations that boosted observers see; the drive's positive-energy characterization rests on its predominantly positive invariant energy density [6]. Its irrotational shift establishes Type I, but does not establish energy-condition satisfaction. The composite diagnostic of Section 3.6 uses a specified slice, wall mask, numerical caps and floors, and affine normalization. Finite-segment null-geodesic and quantum-inequality diagnostics are reported in Section 3.5.

Prior warp-drive tooling (WarpFactory [13]) samples $\sim 10^3$ observer directions per point; we evaluate the Type I eigenvalue criteria and Type-IV labels directly, using curvature computed by automatic differentiation (Section 1.1). Shell constructions such as the regularized WarpShell [8, 9] sit outside the matched benchmark set; recent positive-energy constructions [7, 10] are compared separately in Section 3.6.

1.1 Related work

Energy conditions in warp-drive spacetimes. Hawking and Ellis [3] give the standard algebraic classification of $T^a{}_b$ into four types; Martín-Moruno and Visser give its modern treatment and the conditions under which each energy condition can be violated at each type [5, 14–17]. Santiago, Schuster and Visser [2] proved that within the generic Natário framework physically reasonable warp drives violate the weak, strong and dominant conditions; their null-energy conclusion also requires the subsidiary conditions stated in that work. The qualification "physically reasonable" is theirs, a hypothesis on the localization of the warp field. Their conclusions are observer-independent but not framework-independent, and Barzegar, Buchert and Vigneron [18] examine the routes that would evade its hypotheses. An irrotational shift on time-independent Euclidean slices with unit lapse gives Type-I stress-energy, as in Rodal's construction [6]. Rodal also reports an increase in integrated negative energy when shift vorticity is introduced. That integrated result does not determine the pointwise algebraic type. Olum [19] showed that superluminal travel, under his destination-surface definition and generic-condition hypothesis, requires negative energy. Lobo and Visser [20] established the unavoidable energy-condition violation of the Alcubierre and Natário classes. The applicability of Olum's result requires care: Shoshany and Snodgrass [62] argue that its hypotheses, in particular the generic condition, fail for a wide class including Alcubierre and Van den Broeck. Kontou and Sanders [21] review the modern status of the conditions. The strong condition is violated by a positive cosmological constant, for

which $\rho + 3p = -\Lambda/4\pi < 0$. The Bobrick–Martire taxonomy of *physical warp drives* [8] and the positive-energy construction of Fell and Heisenberg [9] examine how the geometric assumptions restrict the allowed stress-energy.

Warp-drive geometries. The foundational constructions remain Alcubierre's 1994 metric [1], Natário's zero-expansion variant [23] and Van den Broeck's modified-volume construction [24]; the Krasnikov tube [25] replaces the moving bubble with a static corridor. The Lentz soliton [26] was proposed as a positive-energy alternative but was subsequently shown to violate the NEC [2] and the WEC in the Eulerian frame [22]. The Rodal metric [6] completes our benchmark set; Rodal's earlier work [27, 28] analyzes curvature invariants for Alcubierre and Natário in the same $3+1$ formalism, and later notes argue against low-energy warp drives on metamaterial gravitational-coupling grounds [31] and develop photon-propagation branch diagnostics on black-hole backgrounds [58]. Among the 2024–2026 constructions, Fuchs *et al.* [7] introduce a constant-velocity shell-based positive-energy drive in the Bobrick–Martire family; Garattini and Zatrimaylov [29] show that embedding a warp drive on a black-hole background alleviates energy-condition violations, and generalize to a de Sitter background [10]. Their de Sitter analysis expresses the weak and null conditions "up to a total divergence term that averages to zero". These divergence identities do not establish pointwise energy-condition satisfaction. Abellán, Bolívar and Vasilev [57] treat the lapse as an additional degree of freedom, a nontrivial lapse accommodating a heat-flux matter content, complementary to the unit-lapse, flat-slice family analyzed here. Clough, Dietrich and Khan [30] evolve a warp-drive containment failure numerically, complementing the pointwise analysis here, and White *et al.* [36] segment the bubble into interior-flat Gaussian nacelles, reshaping the energy distribution rather than the all-observer character of the wall. Huey [32] proposes a superluminal drive sourced by non-compact thin membranes whose distributional stress-energy satisfies the WEC; such constructions sit outside both the Santiago–Schuster–Visser hypotheses and the smooth-metric scope used here. Santos-Pereira *et al.* [33] study Darmois junction conditions for matching Alcubierre to Minkowski.

Two classification programs bear directly on what is measured here. Barzegar and Buchert [34] identify shared restrictions of the current warp-drive spacetimes (flow-orthogonal foliations, vanishing spatial curvature) and establish, for a one-dimensional shift at $\Lambda = 0$, the relation $E = -\Omega^2/8\pi \leq 0$ between the coordinate vorticity and the Eulerian energy density. Fell and Heisenberg [9] relate vanishing shift vorticity to vanishing Eulerian momentum. Section 3.2 compares momentum and numerical type labels for the sampled geometries. Barzegar, Buchert and Vigneron [18] develop a general classification and no-go results involving global hyperbolicity, asymptotic flatness and positive energy. Under their R-Warp hypotheses a harmonic-gradient shift implies Minkowski; this gives no conclusion for nonharmonic Rodal (Lemma 3). They also question the consistency of the Fuchs shell's assumed matter source with its metric. Our pointwise tests of $G_{ab}/8\pi$ do not resolve that source-consistency question. Buchert and Frackowiak [35] construct new warp solutions and analyze their coordinate-acceleration and vorticity structure kinematically. These geometric classifications complement the Hawking–Ellis classification of the induced stress-energy.

Numerical methods. Computational tools for general relativity span symbolic algebra (xAct, including xPerm [37], and SageManifolds [38]) and numerical-relativity frameworks (the Einstein Toolkit [39], which evolves spacetime geometries); see Ref. [40] for a survey. The pointwise tests combine causal-cone algebra (Theorem 1) with interval arithmetic (Appendix E.1). Bergqvist and Senovilla [61] characterize the tensors with the dominant property as the convex cone generated by superenergy tensors of simple forms, and Martín-Moruno and Visser develop the complementary polynomial route [5, 16]. The S-lemma supplies the pointwise linear matrix inequality used here; the underlying S-lemma and Lorentz-cone results are standard [59, 60, 64–66]. Interval arithmetic with directed rounding encloses numerical errors. Appendix C propagates such enclosures through $g \to \Gamma \to R^a{}_{bcd} \to G_{ab} \to T_{ab}$. A strict bound can establish the sign of an energy-condition contraction; a saturated point can remain inconclusive.

The warp-drive community's dedicated toolkit, WarpFactory, of Helmerich *et al.* [13, 41, 42], evaluates the conditions over a discrete sample of $\sim 10^3$ observer directions and velocities and uses fourth-order central finite differences for the curvature chain. WARPAX uses forward-mode automatic differentiation in JAX [11] for the curvature chain, as in diffjeom [43] and the neural *Einstein Fields* representation [44]. Bara [45] discusses prospective astrophysical applications of automatic differentiation. WARPAX combines the algebraic energy-condition tests with continuous observer optimization [46] and geodesic integration via Diffrax [47], in an Equinox [12] module

structure. Numerical Type-IV detection uses the mixed tensor eigensystem, with the tolerances stated in Section 2.2.

Section 2 presents the methods, Section 3 the energy-condition results, Section 4 the convergence validation, Section 5 the implications and Section 6 the conclusions; the WARPAX implementation (Appendix A) and the metric definitions (Appendix D) are given in the appendices.

1.2 Scope and limitations

The energy-condition criteria are exact algebraic statements. Their numerical evaluation has a separate accuracy limit. Type-I eigenvalue inequalities apply when the type is established; numerical Hawking–Ellis labels depend on eigenvalue and eigenvector tolerances. The LMI criteria of Theorem 1 apply independently of those labels. On the full grids, their binary64 evaluation uses the noise floor (61) and returns *inconclusive* within it.

An exact multiplier or dual certificate establishes a verdict for its stated input tensor. Interval evaluation of the metric and curvature also encloses that input, giving pointwise verdicts and global bounds for the specified wall null deficit (Appendices C and E.1). Sampled type fractions, miss rates, and basin-local extrema retain their numerical status. Their grid stability does not supply a rigorous error bound.

The rapidity-capped observer search measures observer-dependent contractions. A negative contraction supplies a violation witness; a nonnegative search result cannot establish satisfaction for all boosts. Throughout this paper, *observer-robust* refers to the frame-independent truth of the pointwise energy condition. Numerical severities depend on the stated normalization and, for the timelike search, the cap. The single-frame comparison uses the Eulerian normal n^a, which is unit timelike at every speed. We study $v_s \in [0.1, 2.5]$, including the luminal transition.

For the Alcubierre form $g_{00} = -1 + (v_s f)^2$, so ∂_t is null exactly on the locus

$$v_s\, f(r_s) = 1, \tag{6}$$

and spacelike where $v_s f > 1$. Since $f(0) = 1$ and $f'(r_s) < 0$ for every $r_s > 0$, the shape function is strictly decreasing, so (6) has a solution only for $v_s \geq 1$, and at $v_s = 1$ that solution is the single point $r_s = 0$, not a surface and not the wall's inner edge. The locus lies inside the active wall mask $0.1 \leq f \leq 0.9$ precisely when $f = 1/v_s$ falls in that band, that is for

$$10/9 \leq v_s \leq 10, \tag{7}$$

and not for every $v_s > 1$. At this locus ∂_t ceases to define a timelike observer, but the spacetime remains Lorentzian. The determinant is $\det g = -1$ for Alcubierre, Natário, and Rodal, and $\det g = -B^6 < 0$ for Van den Broeck. The Eulerian normal $n^a = (1/\alpha)(\partial_t - \beta^i \partial_i)$ stays unit timelike ($\alpha = 1$, $g^{00} = -1$ at all v_s). The eigenvalues of $T^a{}_b$ are boost invariants and vary continuously as a multiset across the transition. The Hawking–Ellis classification therefore remains well defined. The computed curvature and energy-condition quantities are finite at $v_s \in \{1.0, 1.5, 2.0, 2.5\}$, and we compute the type and eigenvalue structure there (Section 3, Table 3). The continuous optimizer, used only at non-Type I points, parameterizes future-directed unit-timelike vectors of the metric at each point, so its observer domain follows the local light cones through the transition rather than a fixed background cone. We also guard against the principal numerical risk of the superluminal eigenproblem, a spurious Type IV from an ill-conditioned non-symmetric $T^a{}_b$: the Type-IV labels reported here are compared on selected samples using standard and generalized eigensolvers and a 50-digit recomputation. These comparisons do not certify every grid label or convergence of a spatial fraction. What is decided at $v_s > 1$ is the energy-condition verdict, by Theorem 1; the algebraic labels there are checked, not certified. Neither establishes physical realizability or the global causal properties of a superluminal construction.

We compute $T_{ab} = G_{ab}/8\pi$ from prescribed metrics, the Synge G-method. The resulting energy-condition tests characterize the induced classical tensor. They do not establish a realizable matter source, well-posed matter evolution, or global causal viability. These additional requirements are discussed by Barzegar, Buchert, and Vigneron [18].

Pointwise NEC violation does not establish violation of the averaged null energy condition (ANEC) on a complete null geodesic [48]. Section 3.5 reports finite-segment null-energy integrals with selected-ray step refinement. No omitted-tail or validated trajectory-error bound is available. The quantum-inequality comparison is a flat-space estimate [49, 50]; applying a curved-space bound to these worldlines requires a separate treatment [21, 51, 52].

2 Methods

The calculation starts from a prescribed metric and its induced stress-energy. We then test the energy conditions over all causal directions. Algebraic classification helps interpret the tensor, while the matrix criteria provide a test even when the type is difficult to determine numerically. Observer searches show which measurements reveal a violation.

2.1 Stress-energy via automatic differentiation

The Einstein field equations in geometric units ($G = c = 1$) are

$$G_{ab} = 8\pi\, T_{ab}\,, \tag{8}$$

where the Einstein tensor $G_{ab} = R_{ab} - \frac{1}{2} R\, g_{ab}$ is constructed from the Ricci tensor R_{ab} and scalar curvature $R = g^{ab} R_{ab}$. We use the total effective tensor $T_{ab} = G_{ab}/8\pi$; for a cosmological background this includes $-\Lambda g_{ab}/8\pi$. It is not the material tensor obtained after subtracting that contribution. Computing T_{ab} from a given metric requires the full curvature chain:

$$g_{ab} \xrightarrow{\partial} \Gamma^{a}{}_{bc} \xrightarrow{\partial} R^{a}{}_{bcd} \xrightarrow{\text{contract}} R_{ab},\, R \xrightarrow{\text{EFE}} T_{ab}\,. \tag{9}$$

The Christoffel symbols of the second kind are

$$\Gamma^{a}{}_{bc} = \frac{1}{2}\, g^{ad} \left(\partial_c\, g_{bd} + \partial_b\, g_{cd} - \partial_d\, g_{bc}\right)\,, \tag{10}$$

and the Riemann curvature tensor is

$$R^{a}{}_{bcd} = \partial_c\, \Gamma^{a}{}_{bd} - \partial_d\, \Gamma^{a}{}_{bc} + \Gamma^{a}{}_{ce}\, \Gamma^{e}{}_{bd} - \Gamma^{a}{}_{de}\, \Gamma^{e}{}_{bc}\,. \tag{11}$$

WARPAX implements this chain with JAX's forward-mode automatic differentiation. The metric function $g_{ab}(x^\mu)$ maps a coordinate 4-vector to a 4×4 symmetric matrix. Forward-mode differentiation yields $\partial_c\, g_{ab}$ as exact derivatives of the implemented program, up to roundoff, with no step size and no finite-difference truncation error; nesting it (forward-over-forward) gives the second derivatives needed for the Riemann tensor. For the metric map $\mathbb{R}^4 \to \mathbb{R}^{16}$, forward mode requires four input directions, whereas reverse mode requires one pass per scalar output to form the full Jacobian. The derivative convention places the differentiation index last, $(\partial_c\, g)_{abc} = \partial g_{ab}/\partial x^c$, which fixes the index order in the contractions.

Three energy-density quantities. Three energy densities recur throughout this paper. The *invariant energy density* ρ at a Type-I point is minus the timelike eigenvalue of $T^{a}{}_{b}$; it equals $T_{ab} u^a u^b$ on a unit timelike eigenvector. The *observer-dependent energy density* $T_{ab}\, u^a u^b$ is what an observer with four-velocity u^a measures. The *Eulerian energy density* $T_{ab}\, n^a n^b$ is the case of n^a the unit normal to the ADM spatial hypersurface, the single-frame baseline used in this paper.

Notation. R_b denotes the warp bubble radius throughout, distinct from the Ricci scalar R of the curvature expressions.

2.2 Hawking–Ellis classification

The algebraic classification of $T^{a}{}_{b}$ [3] identifies whether a local rest frame exists. When it does, energy-condition tests reduce to inequalities on the rest-frame density and principal pressures. The four types describe the eigenstructure of $T^{a}{}_{b} = g^{ac} T_{cb}$:

- **Type I:** Four real eigenvalues with a timelike eigenvector. This class includes perfect fluids. For signature $(-+++)$ they are $(-\rho, p_1, p_2, p_3)$; the timelike eigenvector is identified by $g_{ab}\, v^a v^b < 0$, with $\rho = -\lambda_{\text{timelike}}$, $p_i = \lambda_{\text{spacelike},i}$.
- **Type II:** A defective 2×2 null Jordan block, a repeated eigenvalue with a single null eigenvector in that block. Canonical example: null dust $T_{ab} = \Phi^2 k_a k_b$ with k^a null [5].
- **Type III:** 3×3 null Jordan structure, a null Jordan block of size three [15].
- **Type IV:** One complex-conjugate eigenvalue pair and two real eigenvalues, with no causal eigenvector. Numerical labels depend on the classification tolerances (Section 1.2).

A Type IV tensor has a negative contraction along some null direction and therefore violates the null energy condition [3, 5, 14]. The violating direction is exhibited in closed form for the momentum-sourced case below. The defective null cases Type II and Type III are *not* decided by that contraction; Appendix E gives tests that also apply to these types.

In an orthonormal frame adapted to the slice normal n, write $\rho_n = T(n,n)$, $j_i = -T(n,e_i)$, and $S_{ij} = T(e_i,e_j)$. For $j \neq 0$, let $\hat{\jmath} = j/|j|$ and $S_\parallel = S(\hat{\jmath},\hat{\jmath})$. Type II can occur on a continuous transition between Types I and IV. Under the splitting hypothesis, $\hat{\jmath}$ is a principal axis of S and the momentum plane decouples. Its block is $A = \left(\begin{smallmatrix} -\rho_n & j \\ -j & S_\parallel \end{smallmatrix}\right)$, which is self-adjoint with respect to $\eta_2 = \mathrm{diag}(-1,1)$; here j denotes the momentum component along $\hat{\jmath}$. With $\Delta = (\rho_n + S_\parallel)^2 - 4j^2$, the type is

$$\Delta > 0 \;\Rightarrow\; \text{I}, \qquad \Delta < 0 \;\Rightarrow\; \text{IV}, \qquad \Delta = 0,\ j \neq 0 \;\Rightarrow\; \text{II}, \qquad \Delta = 0,\ j = 0 \;\Rightarrow\; \text{I (degenerate)}. \tag{12}$$

At $\Delta = 0$ with $j \neq 0$ the repeated eigenvalue $(S_\parallel - \rho_n)/2$ leaves $A - \lambda I$ of rank one with the single null eigendirection $(1, \pm 1)$: a defective null J_2, hence Type II. At $\Delta = 0$ with $j = 0$ the block reduces to $-\rho_n \mathbb{1}$ and stays diagonalizable. Since Δ is continuous, every path from $\Delta > 0$ to $\Delta < 0$ at fixed $j \neq 0$ crosses the Type II locus, so Types I and IV do not exhaust the possibilities. Types II and III are unstable under small perturbations of $T^a{}_b$, while Type IV and nondegenerate Type I are stable [5]. The Type-II locus has codimension one in the momentum-block parameter space away from $j = 0$. Its preimage in a spacetime need not have zero volume, and a finite grid need not sample a transition locus. Type III is excluded under the splitting hypothesis: in a true $2 \oplus 2$ split the momentum block admits Jordan blocks of size at most two and the transverse block is Euclidean-self-adjoint, hence diagonalizable, so no J_3 can be assembled; where the split fails, in the transverse channel of Corollary 1, it is not. An explicit Lorentz-self-adjoint example is $T_{ab} = \ell_a m_b + m_a \ell_b$ with $\ell^a = (1,1,0,0)$ null, $m^a = (0,0,1,0)$ unit spacelike and $\ell \cdot m = 0$, whose mixed form

$$T^a{}_b = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \tag{13}$$

satisfies $(T^a{}_b)^3 = 0$ with $(T^a{}_b)^2 \neq 0$. Its eigenvalues all vanish, its eigenspace is spanned by the null ℓ^a and the spacelike $(0,0,0,1)$, and the Jordan form is $J_3 \oplus J_1$: Segre type [3, 1], Hawking–Ellis Type III. Its Eulerian data are $\rho_n = 0$, $j^a = (0,0,1,0)$, $|j| = 1$ and $S_\parallel = 0$ with $S_{xy} = 1$, so $\hat{\jmath}$ is not a principal axis of S and $\Delta = -4 < 0$: the momentum discriminant would report Type IV at a point that is Type III. The splitting hypothesis cannot be omitted from the Type III exclusion or the $\Delta < 0$ criterion in (12). Gergely's $2+1+1$ analysis of the proper kinetic gravity braiding tensor gives the same discriminant trichotomy for a different stress-energy [4]. The numerical classifier alone does not establish that a sampled near-degenerate point lies on the exact locus.

The Hawking–Ellis label can change across $\Delta = 0$. In a fixed tetrad the multiplier margin $\max_\sigma \lambda_{\min}(\widehat{T} + \sigma\eta)$ is half the normalized null-cone minimum (Corollary 2) and is 1-Lipschitz in the operator norm of $\widehat{T}$: Weyl's eigenvalue inequality bounds the change for each σ, and taking the maximum preserves that bound. Its sign determines the NEC verdict, which can change at zero margin. Theorem 1 applies independently of the type. We use the Type-II convention $\widehat{T} = \left(\begin{smallmatrix} \mu+f & -f \\ -f & -\mu+f \end{smallmatrix}\right) \oplus [p_2] \oplus [p_3]$ in an orthonormal tetrad. For example, $(\mu, f, p_2, p_3) = (0, 1, -2, 0)$ has a positive null part but null-cone minimum $-4/3$ (Appendix E). Positive null dust $T_{ab} = \Phi k_a k_b$, $\Phi > 0$, instead satisfies all four conditions: $T(u,u) = \Phi(k \cdot u)^2 \geq 0$, its trace vanishes, and its flux is future null or zero.

The mixed tensor $T^a{}_b = g^{ac}T_{cb}$ is generally non-symmetric, so WARPAX uses a general eigensolver; eigenvalues count as real when $|\mathrm{Im}\,\lambda_i| < \epsilon s$ ($\epsilon = 10^{-10}$, $s = \max(|\mathrm{Re}\,\lambda_j|, \|T^a{}_b\|_\infty)$) (Section 1.2 compares selected Type-IV labels using three solver approaches). Scaling the tolerance with the tensor preserves the test under $T \to cT$ for nonzero c. A unit floor would instead impose an absolute cutoff on weak stress tensors and could suppress a nonreal eigenvalue pair. The vacuum shortcut requires $T^a{}_b = 0$ and assigns Type I with $\rho = p_i = 0$. A small real part of the spectrum is insufficient: pure momentum flux has eigenvalues $\pm iq$, and a nonzero nilpotent tensor has an entirely zero spectrum. The separate WarpShell diagnostic in Appendix E includes near-vacuum labels, which are counted separately there. For Type I, all energy conditions reduce to

algebraic inequalities in the rest-frame density ρ and principal pressures p_1, p_2, p_3:

$$\text{NEC:} \quad \rho + p_i \geq 0 \qquad \forall\, i\,, \tag{14}$$

$$\text{WEC:} \quad \rho \geq 0 \text{ and } \rho + p_i \geq 0 \qquad \forall\, i\,, \tag{15}$$

$$\text{SEC:} \quad \rho + p_i \geq 0 \;\; \forall\, i \;\text{ and }\; \rho + \textstyle\sum_i p_i \geq 0\,, \tag{16}$$

$$\text{DEC:} \quad \rho \geq |p_i| \qquad \forall\, i\,. \tag{17}$$

These inequalities are *exact*: for Type I stress-energy a point satisfies (or violates) an energy condition if and only if the corresponding eigenvalue inequality holds, whichever observer is chosen. The classification decision itself is tolerance-based in floating point (Section 1.2), so the numerical verdict depends on eigenvalue accuracy, degeneracy tolerances, and the timelike eigenvector test.

Deciding outside Type I. Decompose T in the Eulerian frame into the Eulerian energy density $\rho_n = T_{ab}\, n^a n^b$ (equal to the Type-I rest-frame density ρ when the momentum vanishes), the momentum density $j^i = -h^i{}_a\, T^a{}_b\, n^b$ and the spatial stress $S_{ij} = h_i{}^a h_j{}^b\, T_{ab}$, with $h^a{}_b = \delta^a{}_b + n^a n_b$ the projector onto the slice. In the timelike plane spanned by n^a and the momentum direction $\hat{\jmath}^a$ the mixed tensor is the block $\left(\begin{smallmatrix} -\rho_n & |j| \\ -|j| & S_\parallel \end{smallmatrix}\right)$, $S_\parallel = S(\hat{\jmath}, \hat{\jmath})$, whose eigenvalues $\frac{1}{2}(S_\parallel - \rho_n) \pm \frac{1}{2}\sqrt{\Delta}$ are complex exactly when the momentum density dominates,

$$\Delta = (\rho_n + S_\parallel)^2 - 4|j|^2 < 0, \tag{18}$$

that is $2|j| > |\rho_n + S_\parallel|$. Such a pair makes the point Type IV when $\hat{\jmath}^a$ is a principal axis of S_{ij}, and one of the Eulerian null vectors $k^a = n^a \pm \hat{\jmath}^a$ then witnesses the violation,

$$T_{ab}\, k^a k^b = \rho_n + S_\parallel - 2|j| < 0, \tag{19}$$

with no rapidity and no optimizer. A Type-IV point whose complex pair lies outside that plane, for example one opened by a conformal spatial factor, still has no causal eigenvector, so the classification guarantees its null-energy violation even where this witness is nonnegative. At Type II a nonnegative value of the same contraction does not establish the NEC: it probes the momentum plane only, leaving a transverse violation undetected, and on the canonical block $(\mu, f, p_2, p_3) = (0, 1, -2, 0)$ it is exactly zero while the null-cone minimum is $-4/3$. Type II is therefore decided by Appendix E, not by (19). The discriminant (18) is compared with the full eigensolver in Section 3.2.

Observer interpretation of the Type-I criteria. Let $\{e_0^a, e_i^a\}$ be an orthonormal eigenframe at a Type-I point, with $T(e_0, e_0) = \rho$ and $T(e_i, e_j) = p_i \delta_{ij}$. For $u^a = \cosh\zeta\, e_0^a + \sinh\zeta\, \hat{s}^a$,

$$T(u,u) = \rho + \sinh^2\zeta\, [\rho + p(\hat{s})], \qquad p(\hat{s}) = \sum_i p_i (\hat{s}\cdot e_i)^2.$$

Nonnegativity for every rapidity and direction requires and suffices for $\rho \geq 0$ and $\rho + p_i \geq 0$, the WEC criteria. For a normalized null vector $k = e_0 + \hat{s}$, $T(k,k) = \rho + p(\hat{s})$; the NEC therefore requires no separate sign condition on ρ. Applying the timelike test to $T_{ab} - \frac{1}{2} T g_{ab}$ gives the SEC, while future-causal energy flux gives $\rho \geq |p_i|$ for the DEC [3, 14]. The rest frame e_0 need not coincide with the Eulerian normal n.

The momentum discriminant. The complex pair of (18) has imaginary part $\frac{1}{2}\sqrt{-\Delta}$.

At a point with $j = 0$, $T^a{}_b$ is block diagonal, $(-\rho) \oplus S$ with S symmetric, so its spectrum is real and the point is Type I. For a flat-slice unit-lapse drive whose shift is $\beta^i = v_s F^i(\mathbf{x} - v_s t \mathbf{e}_x)$ with fixed smooth F, the scalings $\rho_n, S_{ij} \propto v_s^2$ and $|j| \propto v_s$ are exact for the Einstein tensor rather than leading order, as follows from the $3+1$ equations. The extrinsic curvature $K_{ij} = \partial_{(i}\beta_{j)}$ is linear in v_s; on a flat slice the Hamiltonian constraint $16\pi\rho_n = K^2 - K_{ij}K^{ij}$ is quadratic in K, and the momentum constraint $8\pi j^i = D_j(K^{ij} - \gamma^{ij}K)$ is linear in it. In the evolution equation for K_{ij}, with $\alpha = 1$ and $\gamma_{ij} = \delta_{ij}$, every surviving term is quadratic in K or is $\pounds_\beta K$, while $\partial_t K_{ij} = -v_s \partial_x K_{ij}$, so $S_{ij} - \frac{1}{2}\delta_{ij}(S - \rho_n)$ is quadratic as well and with ρ_n so is S_{ij}. When $j = 0$, this also implies $\min_i(\rho + p_i) = -C v_s^2$ at a fixed comoving point, or after taking an infimum over a fixed comoving wall domain. Here C is independent of speed; $C > 0$ requires a null violation in that domain. With $j \neq 0$, diagonalization also involves the speed-linear momentum terms and the quadratic margin law does not follow. Under the splitting hypothesis, a complex pair occurs wherever the

momentum density dominates, $2|j| > |\rho_n + S_\parallel|$; writing $|j| = d\,v_s$ and $\rho_n + S_\parallel = a\,v_s^2$ this is $\Delta = v_s^2(a^2 v_s^2 - 4d^2) < 0$, so at a point with $d > 0$, $a \neq 0$ and $v_s > 0$ the momentum-plane pair closes at the exact speed $v_s = 2d/|a|$, above which the decoupled tensor is Type I (Section 3.1). The null-witness deficit $\rho_n + S_\parallel - 2|j|$ then has a linear-in-$|j|$ term absent in the irrotational case.

Because a and d are independent of v_s, the field $v_\star(\mathbf{x}) = 2d/|a|$ is a property of the fixed shift profile. Where $a \neq 0$ the momentum channel is open for $v_s < v_\star(\mathbf{x})$ and closed above it; where $a = 0$ with $d > 0$ it never closes. For $d = 0$ the momentum channel is absent, and we set $v_\star = 0$. Its predicted volume fraction is

$$F_{\mathrm{mom}}(v_s) = \frac{\mathrm{vol}\{\, v_\star > v_s \,\}}{\mathrm{vol(wall)}}. \tag{20}$$

F_{mom} bounds the Type-IV fraction only under the splitting hypothesis, with $\hat{\jmath}^a$ a principal axis of S_{ij}: $\Delta < 0$ is sufficient for Type IV under that hypothesis and not otherwise, since switching on the transverse coupling of Corollary 1 can restore a real spectrum to the momentum pair. On the block of that corollary the cubic discriminant is even in m with leading coefficient $+4m^6$, so at fixed $\Delta < 0$ and nonvanishing minor there is a finite $|m|$ past which $D > 0$: three real eigenvalues, Type I, with $\Delta < 0$. At $(\rho_n, S_x, S_y, |j|) = (1, 1, 0, 2)$, where $\Delta = -12$, the discriminant is negative at $m = 3$ and positive at $m = 4$. Thus $F_{\mathrm{mom}} \leq F_{\mathrm{IV}}$ requires the splitting premise. Section 3.3 compares aggregate numerical fractions; that comparison does not check the premise pointwise.

Corollary 1 (Type-IV boundary with transverse stress coupling)**.** *Let the momentum direction $\hat{\jmath}^a$ fail to be a principal axis of the spatial stress S_{ij}, so a transverse coupling m enters the block along n^a, $\hat{\jmath}^a$, and the coupled transverse axis $\hat{t}^a$; and let the remaining spatial axis $\hat{z}^a$ decouple from that block, $S(\hat{\jmath}, \hat{z}) = S(\hat{t}, \hat{z}) = 0$. The decoupling is a hypothesis and not a normalization: without it the characteristic polynomial of $T^a{}_b$ is quartic and the cubic discriminant of the block does not decide the type. Its cubic discriminant $D(\rho_n, S_x, S_y, |j|, m)$ is even in m and satisfies $D|_{m=0} = (\mathrm{minor})^2\,\Delta$, so a transverse complex pair (Type IV) opens where $D < 0$, its boundary lying on the algebraic surface $D = 0$, and it can open with $\Delta \geq 0$: the momentum discriminant of Section 2.2 is sufficient under the splitting hypothesis, but is not necessary for Type IV without that hypothesis. On the witness family $\rho_n = S_x = 1$, $|j| = \frac{1}{2}$, $S_y = 0$ the transverse pair is complex on the band $m_\star < |m| < m_2$ between the two positive roots of D, with $m_\star \approx 0.782$ and $m_2 \approx 2.015$, and at $m = 1$ an Eulerian null vector of the orthonormal frame exhibits the null violation directly.*

The witness family is a representative stress ratio chosen to exhibit the transverse channel, not a sampled Van den Broeck wall point; the overall scale is irrelevant to the type.

Proof. In the orthonormal Eulerian frame $(n^a, \hat{\jmath}^a, \hat{t}^a, \hat{z}^a)$, with $\hat{t}^a$ the transverse axis coupled to $\hat{\jmath}^a$ by $m = S(\hat{\jmath}, \hat{t})$ and $\hat{z}^a$ decoupled (real eigenvalue S_z), the mixed tensor on the $(n, \hat{\jmath}, \hat{t})$ block is

$$T^a{}_b\big|_{\mathrm{block}} = \begin{pmatrix} -\rho_n & |j| & 0 \\ -|j| & S_x & m \\ 0 & m & S_y \end{pmatrix}, \qquad S_x = S_\parallel.$$

Its characteristic polynomial is $\lambda^3 + c_2\lambda^2 + c_1\lambda + c_0$, with

$$\begin{aligned} c_2 &= \rho_n - S_x - S_y, \\ c_1 &= S_x S_y - m^2 - \rho_n(S_x + S_y) + |j|^2, \\ c_0 &= \rho_n(S_x S_y - m^2) - |j|^2 S_y. \end{aligned}$$

The cubic discriminant is

$$D = 18c_2c_1c_0 - 4c_2^3c_0 + c_2^2c_1^2 - 4c_1^3 - 27c_0^2,$$

a polynomial even in m. The surface $D = 0$ also contains real-eigenvalue degeneracies away from the Type-I/Type-IV transition. Setting $m = 0$ gives $D|_{m=0} = (\mathrm{minor})^2\,\Delta$ with $\mathrm{minor} = S_xS_y - S_y^2 - |j|^2 + (S_x - S_y)\rho_n$ and $\Delta = (\rho_n + S_x)^2 - 4|j|^2$, recovering Section 2.2 on a principal axis. For the witness family $\rho_n = S_x = 1$, $|j| = \frac{1}{2}$, $S_y = 0$, the discriminant becomes, in $u = m^2$, $D = 4u^3 - 18u^2 + \frac{27}{4}u + \frac{27}{16}$, whose two positive roots $u_\star \approx 0.611$ and $u_2 \approx 4.059$ bound the Type-IV band $m_\star = \sqrt{u_\star} \approx 0.782 < |m| < \sqrt{u_2} = m_2 \approx 2.015$; for m in this band the block carries a complex pair, and at $m = 1$ the null vector $k^a = n^a + (\hat{\jmath}^a - \hat{t}^a)/\sqrt{2}$ gives $T_{ab}\,k^a k^b = \rho_n + \frac{1}{2}S_x + \frac{1}{2}S_y - \sqrt{2}\,|j| - m = \frac{1}{2} - \frac{1}{\sqrt{2}} < 0$. □

The momentum and transverse reductions require their respective decoupling hypotheses. Table 4(b) compares the measured imaginary part with $\frac{1}{2}\sqrt{\max(0,-\Delta)}$ at a selected off-axis point; agreement or disagreement at that point does not establish a wall-wide mechanism. The numerical type labels use the full four-dimensional tensor. For Garattini–Zatrimaylov, the type is established analytically as Type I (Section 3.6).

The algebraic slack, $\min_i(\rho + p_i)$ and its WEC/SEC/DEC analogs, is the smallest left-hand side of the corresponding Type-I inequalities. Its sign determines satisfaction, and its magnitude measures proximity to the boundary in the eigenframe. The rapidity-capped minimum $\min_{\zeta\le\zeta_{\max}} T_{ab}\,u^a u^b$ depends on the chosen normal and cap. A negative value witnesses WEC violation; a nonnegative capped value does not prove WEC. At a NEC-violating point the unrestricted timelike density tends to $-\infty$ along suitable boosts.

2.3 Pointwise matrix criteria

The energy conditions are inequalities on causal vectors, so they can be tested without first assigning an algebraic type. Fixing the time component in an orthonormal frame reduces the causal directions to a sphere or ball. The S-lemma, a standard result for quadratic inequalities, replaces the test over this set by a matrix inequality with one scalar multiplier. Appendix E gives the proof and explicit certificates.

Let $\{n^a, e_i^a\}$ be an orthonormal tetrad with n^a the unit slice normal, and decompose the stress-energy in it as $\rho_n = T_{ab}n^a n^b$, $b_i = -T_{ab}n^a e_i^b$, $S_{ij} = T_{ab}e_i^a e_j^b$; thus $b_i = j_i$ in this frame. Every future unit timelike u^a is $u^a = \gamma(n^a + w^i e_i^a)$ with $|w| < 1$ and $\gamma = (1-|w|^2)^{-1/2}$, and every future nonzero null k^a is $k^a = c\,(n^a + s^i e_i^a)$ with $|s| = 1$, $c > 0$. Then

$$T_{ab}u^a u^b = \gamma^2 q(w), \qquad q(w) = \rho_n - 2\,b\cdot w + w^{\mathsf{T}} S w, \tag{21}$$

with $\gamma^2 > 0$. This positive prefactor leaves the sign unchanged. The conditions therefore reduce to quadratic inequalities on compact sets: the closed unit ball for the timelike conditions, by continuity, and the unit sphere for the null condition. The S-lemma decides these inequalities without a Hawking–Ellis classification.

Write $\eta = \mathrm{diag}(-1,1,1,1)$ for the tetrad-frame metric and $\widehat{T}$ for the tetrad-frame component matrix of T_{ab}, so that $\widehat{T}_{00} = \rho_n$, $\widehat{T}_{0i} = -b_i$, $\widehat{T}_{ij} = S_{ij}$. Define the one-parameter family

$$M(\sigma) \;=\; \widehat{T} + \sigma\,\eta \;=\; \begin{pmatrix} \rho_n - \sigma & -b^{\mathsf{T}} \\ -b & S + \sigma\,\mathbb{1} \end{pmatrix}, \tag{22}$$

the tetrad-frame matrix of $T_{ab} + \sigma g_{ab}$. The notation $M \succeq 0$ means positive semidefinite: $z^{\mathsf{T}} M z \ge 0$ for every real vector z.

Theorem 1 (Decision by a linear matrix inequality)**.** *Let T_{ab} be a real symmetric tensor at a point of a four-dimensional Lorentzian spacetime. Define $\Theta_{ab} = T_{ab} - \frac{1}{2}T g_{ab}$ with $T = g^{ab}T_{ab}$, and let $(T^2)_{ab} = T_{ac}g^{cd}T_{db}$. Then, with no hypothesis on the Hawking–Ellis type of $T^a{}_b$,*

1. *the NEC holds at the point $\iff$ $M(\sigma) \succeq 0$ for some $\sigma \in \mathbb{R}$;*
2. *the WEC holds $\iff$ $M(\sigma) \succeq 0$ for some $\sigma \ge 0$;*
3. *the SEC holds $\iff$ the WEC holds for Θ_{ab}, i.e. $\iff$ $\widehat{\Theta} + \sigma\eta \succeq 0$ for some $\sigma \ge 0$;*
4. *the DEC holds $\iff$ the WEC holds* and *the WEC holds for $-(T^2)_{ab}$.*

At Type I points we use the eigenvalue inequalities (14)–(17), which are already necessary and sufficient there. At other types we use Theorem 1; a negative Eulerian null contraction (19) also gives a direct violation witness. Where the metric and curvature are evaluated with interval arithmetic, these tests include bounds on the numerical error in T_{ab} (Appendix E.1).

2.4 Observer parameterization

To display observer-dependent contractions, we boost the Eulerian normal for timelike vectors and parameterize the celestial sphere for null vectors. The LMI criterion itself requires no observer search.

Timelike observers. Any future-directed unit timelike vector u^a at a point can be written as a Lorentz boost of the Eulerian normal n^a:

$$u^a = \cosh\zeta\, n^a + \sinh\zeta\, \hat{s}^a(\theta,\phi)\,, \tag{23}$$

where $\zeta \geq 0$ is the rapidity (boost magnitude), (θ,ϕ) specify the spatial boost direction, and $\hat{s}^a(\theta,\phi)$ is a unit spatial vector constructed from an orthonormal tetrad adapted to the ADM decomposition. The tetrad is built by applying Gram–Schmidt orthogonalization to the coordinate basis vectors projected into the spatial hypersurface orthogonal to n^a.

Null directions. A future-directed null direction has the normalized representative

$$k^a = n^a + \hat{s}^a(\theta,\phi)\,, \tag{24}$$

where (θ,ϕ) parameterize the null direction on the celestial sphere. This fixes $k_a n^a = -1$, removing the null-vector rescaling ambiguity. The resulting NEC margin $T_{ab}\,k^a k^b$ is defined relative to n^a. Its minimum is a two-dimensional optimization problem over (θ,ϕ).

In the rapidity parameterization, $\zeta = 0$ recovers the Eulerian observer, so the optimization always includes the Eulerian result as a baseline. The WEC, SEC, and DEC involve three-dimensional optimization over (ζ,θ,ϕ), while the NEC requires only (θ,ϕ). The true timelike observer manifold has unbounded rapidity ($\zeta \in [0,\infty)$); in practice, we optimize over $\zeta \in [0,\zeta_{\max}]$ with a fixed cap $\zeta_{\max} = 5$ (corresponding to $\gamma \approx 74$; see Section 2.5).

2.5 Observer optimization

The observer diagnostic searches for the minimum

$$\text{margin}^* = \min_{(\zeta,\theta,\phi)\in\mathcal{O}} m(T_{ab}, u^a(\zeta,\theta,\phi))\,, \tag{25}$$

where m is the signed margin function: $m \geq 0$ means the condition holds for the tested vector, and $m < 0$ indicates a violation. For the NEC, $m = T_{ab}\,k^a k^b$ with optimization over (θ,ϕ) only. For the DEC, the diagnostic at each spacetime point returns

$$m_{\text{DEC}} = \min\big(m_{\text{flux}},\ m_{\text{future}},\ m_{\text{WEC}}\big)\,, \tag{26}$$

where $m_{\text{flux}} = -g_{ab}\,J^a J^b$ (nonnegative when the energy flux $J^a = -T^a{}_b\,u^b$ is causal), $m_{\text{future}} = -J_a n^a$ (positive when the causal J^a is future-directed, with n^a the future-pointing unit normal), and $m_{\text{WEC}} = T_{ab}\,u^a u^b$ (nonnegative when the energy density is nonnegative). A negative value of any component indicates a DEC violation for that observer. The three sub-terms have mixed dimensions, so we use them only for sign-based detection ($m^*_{\text{DEC}} < 0$), not as a combined severity. For Type I stress-energy the algebraic condition (equation 17) guarantees J^a causal and future-directed for every future-directed u^a, so the algebraic slack $\min_i(\rho - |p_i|)$ is the primary DEC result; the three-term optimizer diagnostic is used only at the residual non-Type I points.

Optimization uses BFGS [53] in Optimistix [46] over an unconstrained boost vector $\mathbf{z} \in \mathbb{R}^3$, with direction $\mathbf{z}/|\mathbf{z}|$ and rapidity $\zeta = \zeta_{\max}\tanh(|\mathbf{z}|/\zeta_{\max})$. At $\mathbf{z} = 0$ the observer is Eulerian. The soft cap is $\zeta_{\max} = 5$ ($\gamma \approx 74$); null directions use a stereographic parameterization of S^2. Multi-start (N_{starts} seeds: Eulerian, six axis-aligned, and random) is evaluated in parallel, batched over the starts.

Closed-form worst observer at Type I points. At a Type I point the rest-frame minimum has a closed form. In the eigenframe $T^a{}_b = \text{diag}(-\rho, p_1, p_2, p_3)$, an observer boosted from the rest frame with rapidity ζ along principal axis i measures

$$\rho_{\text{obs}}(\zeta) = \rho\cosh^2\zeta + p_i\sinh^2\zeta = \rho + (\rho + p_i)\sinh^2\zeta\,, \tag{27}$$

monotone in $\sinh^2\zeta$. If $\rho + p_i \geq 0$ for every axis the rest frame already measures the least energy density; if $\rho + p_{i^*} < 0$ for $i^* = \arg\min_i(\rho + p_i)$, the worst boost lies along the principal eigenvector e_{i^*} of the most-violating pressure, every observer boosted along that axis past the threshold rapidity $\sinh^2\zeta_{\text{th}} = \rho/|\rho + p_{i^*}|$ (for $\rho > 0$) measures negative energy density, and ρ_{obs} is unbounded below as $\zeta \to \infty$. The closed form therefore supplies the worst direction, the violation threshold, and the asymptotic sign; no finite observer attains the unrestricted infimum. Here ζ is measured from the Type-I rest frame. It agrees with the Eulerian rapidity used by the numerical cap only when that rest frame coincides with n^a.

Table 1. *(a)* Per-metric parameters and grid configuration; all grids are 50^3 at $t = 0$, centered on the bubble. This table lists native per-metric parameters for the uniform-grid diagnostics; the primary cross-metric comparison instead evaluates every retained metric at common parameters ($R_b = 1$, $\sigma = 8$) on smoothly graded compact grids (Section 3.1). †Additional Van den Broeck parameters: $\tilde{R} = 1$, $\alpha_{\rm vdb} = 0.5$, $\sigma_B = 8$. *(b)* Wall resolution analysis on the same grids. Transition width is the 10–90% threshold distance of each metric's shape function, root-solved from the profile rather than taken from the large-σR_b limit $\Delta r = 2\,\mathrm{atanh}(0.8)/\sigma$ (72.53 against 73.24 at the native Rodal parameters). The *Convergence* column records the resolution-support tier on the native uniform grid (Stability-only: minimum margin stable under refinement; – : no study).

(a)

Metric	R_b	σ	Domain	$\zeta_{\max}$
Alcubierre	1	8.0	$(\pm5)^3$	5
Van den Broeck†	1	8.0	$(\pm5)^3$	5
Natário	1	8.0	$(\pm5)^3$	5
Rodal	100	0.03	$(\pm300)^3$	5
Schwarzschild	$M=1$	–	$(\pm20)^3$	5

(b)

Metric	Wall width	Δx	Cells	Resolved	Convergence
Alcubierre	0.2747	0.2041	1.35	No	Stability-only
Natário	0.2747	0.2041	1.35	No	–
Van Den Broeck	0.2747	0.2041	1.35	No	–
Rodal	72.5335	12.2449	5.92	Yes	Stability-only
Schwarzschild	–	0.82	–	–	–

2.6 Geodesic integration

WARPAX integrates the geodesic equation

$$\frac{\mathrm{d}^2 x^\mu}{\mathrm{d}\lambda^2} + \Gamma^\mu{}_{\alpha\beta}\,\frac{\mathrm{d}x^\alpha}{\mathrm{d}\lambda}\,\frac{\mathrm{d}x^\beta}{\mathrm{d}\lambda} = 0 \tag{28}$$

with either of two schemes. The toolkit provides the Tsitouras 5(4) Runge–Kutta method (Tsit5) [54] via Diffrax [47], with adaptive step-size control (PID controller, rtol = atol = 10^{-10}). Every geodesic finite-segment value reported here uses the extended-phase-space symplectic scheme of Section 3.5, with a fourth-order Yoshida composition. A numerical tangent can drift from the null cone; these runs monitor that drift and step-refine each selected ray. The coordinate-path diagnostic samples an explicitly prescribed path and does not solve the geodesic equation.

3 Results: Energy conditions

We evaluate energy conditions on 50^3 spatial grids for each metric, with the time coordinate fixed at $t = 0$ (for constant-velocity bubbles whose metric depends on $x - v_s t$, the $t = 0$ slice is representative up to spatial translation). The grid extends $\pm5R_b$ for metrics with $R_b = 1$ (Alcubierre, Van den Broeck, Natário) and $\pm3R_b$ for the large-radius Rodal metric ($R_b = 100$), centered on the bubble.

Automatic differentiation evaluates derivatives of the implemented metric without finite-difference truncation error. Roundoff and profile regularization still affect the computed tensor. Spatial grid refinement tests the sampling of wall extrema and thresholded fractions; geodesic step refinement is a separate test.

On the uniform grids above the Alcubierre/VdB/Natário walls span only ∼1.35 cells (10–90% width $\Delta r \approx 2.2/\sigma$; Table 1(b)). The central benchmark (Section 3.1) therefore evaluates all four metrics at matched parameters ($R_b = 1$, $\sigma = 8$) on one family of wall-clustered graded grids that resolve the 10–90% wall by 5.9 to 8.9 cells on a radial crossing ($\Delta r/\Delta x_{\rm wall}$), a factor of 4.4 to 6.6 over the 1.35 cells of the uniform grid: box $\pm3R_b$, clustering strength $a_\star = 2$, three levels $N = 80, 100, 120$ giving 5.9, 7.6, 8.9 cells at wall spacings $\Delta x_{\rm wall} = 0.0465, 0.0362, 0.0309$, every level clearing the four-cell resolution criterion. (The spacing is the *clustered* one at the wall, not the nominal $6R_b/N$.) We refine sampled wall extrema off-grid within their selected basins and report the grid dependence of the thresholded volume fractions (Section 4). These calculations measure numerical stability; the global enclosures are obtained separately. Table 1(a) lists the per-metric native parameters; unless a caption states otherwise, every figure and table below uses them, and matched-parameter results use $R_b = 1$, $\sigma = 8$, $v_s = 0.5$ on the wall-resolved graded grids just described.

A wall spanning fewer than four cells *on a uniform grid* is excluded from the uniform-grid comparison (this removes the native thin-shell and soliton cases); the matched benchmark instead resolves the wall on the graded grids above, whose observed resolution dependence is reported in Section 4. Cell counts are quoted as the *worst* case over every crossing the axis makes of the wall,

Table 2. Matched-parameter ($R_b = 1$, $\sigma = 8$, $v_s = 0.5$), wall-restricted, proper-volume-weighted invariant benchmark on the wall-resolved graded grid ($N = 100$, ∼7.6 cells across the 10-90% wall). The invariant peak NEC deficit $\min(\rho + p_i)$ is over Type I points, refined off-grid; for the Type-IV-dominated walls it samples only the residual real-eigenvalue points and is not a severity bound. "Missed by Eulerian" is the fraction of all-observer violations at which the Eulerian frame reports none, at $v_s = 0.5$.

	Type I	Type IV	$\min(\rho + p_i)$	Missed by Eulerian (%)		
Metric	(%)	(%)	(Type I)	WEC	NEC	DEC
Alcubierre	1.3	98.7	-0.158	0.0	0.0	0.0
Natário	16.6	83.4	-1.955	0.0	0.0	0.0
Van den Broeck	18.9	81.1	-0.521	54.8	14.1	51.3
Rodal	100.0	0.0	-0.194	72.6	13.2	74.0

measured on the grid actually used: a closed wall is crossed twice per axis, and reporting the better crossing would overstate the resolution.

For each grid point we compute the Eulerian baseline: the ADM normal for WEC/SEC/DEC and six axis-aligned null directions for NEC. The all-observer criteria are the Type-I eigenvalue inequalities and, at other types, the LMI tests of Theorem 1. A negative axial null contraction (19) supplies a direct violation witness. All numerical verdicts remain subject to the accuracy limits of Section 1.2. Away from the wall the stress can be small without being Type I. An unrestricted grid fraction also depends on the size of the exterior box; the primary cross-metric comparisons therefore use wall-restricted fractions (Section 3.1).

3.1 Energy conditions across the luminal transition

The matched-parameter benchmark ($R_b = 1$, $\sigma = 8$) compares the all-observer energy conditions and algebraic types on graded bubble-wall grids across the luminal transition. Exact Hawking–Ellis types and Type-I eigenvalue slacks are Lorentz invariants; their numerical estimates retain the accuracy limits of Section 1.2. Their volume fractions refer to the specified constant-t slice and wall mask. None requires ∂_t to be timelike, so the analysis is identically defined below, at, and above $v_s = 1$.

Wall algebraic types. Table 2 restricts to the active wall ($f \in [0.1, 0.9]$, proper-volume weighted). The Alcubierre, Natário, and Van den Broeck walls are *Type-IV dominated* (81–99% at $v_s = 0.5$), with Type-I fractions 1.3%, 16.6%, 18.9%: an established Type-IV tensor has no timelike eigenvector and violates all four energy conditions [5]. The Rodal samples are 100% Type I, consistent with the exact result for its ideal profile. Their energy-condition tests use the Type-I eigenvalue inequalities. The wall type fractions come from the standalone benchmark grid; their resolution dependence is reported in Section 4.

The all-observer reading of the positive-energy result. Rodal recently put forward a warp drive with predominantly positive invariant energy density and global Type I structure, reported at $v/c = 1$ [6]. Lemma 3 derives that structure for the analytic profile. The sampled comparison illustrates how observer choice enters: the Eulerian frame reports no violation at ≈73% of the wall points where a boosted observer sees a weak-energy violation, and at ≈74% for the dominant energy condition (Table 2). The reference verdicts use the Type-I eigenvalue inequalities. The fractions are sampled and proper-volume weighted, with a range of about two percentage points across the resolution ladder. Global Type I fixes the algebraic character, not energy-condition satisfaction: the irrotational wall still has a residual null (hence weak, dominant, and strong) violation, reduced but nonzero in Rodal's own construction [6] and consistent with the drive-by-drive NEC violation we report below. For the ideal irrotational Rodal profile, $\rho_n = \rho$ because $j = 0$. This equality of densities does not equate a single-observer test with the all-observer energy conditions. At a Type-IV point there is no timelike eigenvector and hence no rest-frame energy density.

Type and severity across $v_s = 1$. Table 3 and Figure 1 track the wall type structure from deep subluminal to superluminal. Across the sampled transition, Rodal remains 100% Type I to $v_s = 2.5$, while the Alcubierre and Natário walls remain Type-IV dominated (Van den Broeck only above its transition, below). The Type-IV fraction decreases overall with speed for Alcubierre and Natário, with small sampled fluctuations. The wall Type I fraction reaches 24.9% (Alcubierre), 43.8% (Natário), and 23.2% (Van den Broeck) at $v_s = 2.5$ (Figure 1), even as the sampled Type-I NEC slacks deepen. Van den Broeck initially shows the opposite trend, passing through a velocity-driven Type-I → Type-IV transition that takes its wall Type-IV fraction from 7.1% at $v_s = 0.1$ to a peak

Table 3. Wall-restricted Hawking–Ellis Type I / Type IV fractions (%, proper-volume weighted) at subluminal, luminal, and superluminal speeds. Computed frame-independently from $T^a{}_b$ on the wall-resolved benchmark grid ($N = 100$, wall resolved to 7.6 cells); Type-IV labels cross-checked (Section 1.2); the full sweep to $v_s = 2.5$ is shown in Figure 1.

	$v_s = 0.5$ (sub)		$v_s = 1.0$ (luminal)		$v_s = 2.0$ (super)	
Metric	Type I	Type IV	Type I	Type IV	Type I	Type IV
Alcubierre	1.3	98.7	5.7	94.3	16.7	83.3
Natário	16.6	83.4	28.4	71.6	41.3	58.7
Van den Broeck	18.9	81.1	21.0	79.0	25.4	74.6
Rodal	100.0	0.0	100.0	0.0	100.0	0.0

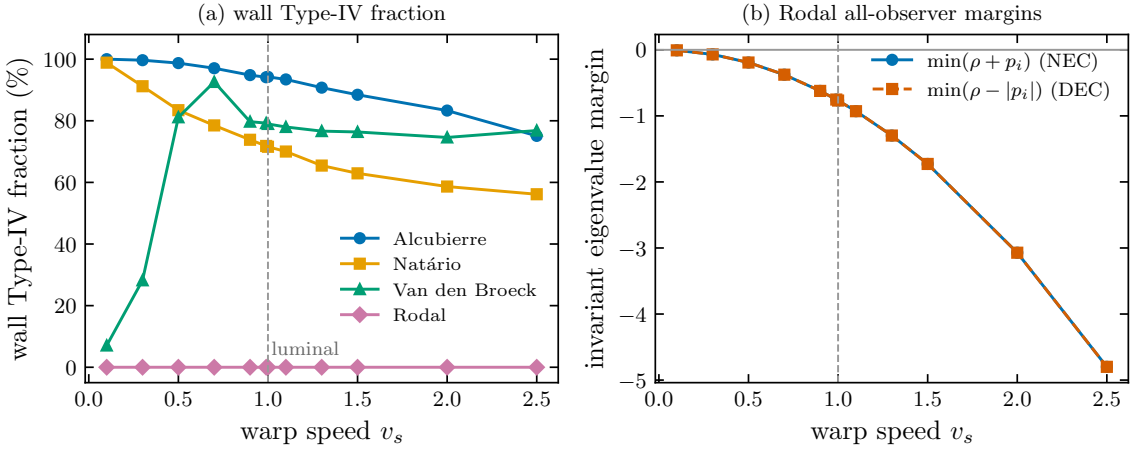

Figure 1. Type structure and energy-condition slacks at $R_b = 1$, $\sigma = 8$ on the $N = 100$ wall grid; $v_s = 1$ is dashed. (a) Proper-volume-weighted Type-IV fractions on $t = 0$, $f \in [0.1, 0.9]$. These aggregate trends do not test the local splitting hypotheses. Rodal's ideal profile is Type I by Lemma 3. (b) Rodal's sampled NEC and DEC minima coincide at the displayed precision and follow the quadratic law for its ideal irrotational profile (equation 39). Type I does not imply energy-condition satisfaction. Lines connect sampled speeds.

of 92.6% at $v_s = 0.7$, its wall Type-IV fraction crossing 50% at $v_s \approx 0.38$; the decline sets in only past that peak. For the everywhere-Type-I Rodal geometry the invariant NEC and DEC margins (Figure 1b) deepen monotonically from $\min(\rho + p_i) \approx -0.0077$ at $v_s = 0.1$ to -4.799 at $v_s = 2.5$, following the quadratic law $\min(\rho + p_i) = -0.768\, v_s^2$ ($R^2 = 1$ over the full sweep $v_s \in [0.1, 2.5]$; the subluminal-only fit of Table 11(b) returns the same coefficient to the displayed precision, $|\min(\rho + p_i)|/v_s^2 = 0.7679023$ at every sampled speed). For the ideal Rodal profile, irrotationality gives $j = 0$ and the quadratic law holds exactly. The separately regularized implementation has the small residual curl discussed in Appendix B; its sampled agreement with that law is numerical evidence. The strictly negative coefficient is consistent with the Santiago–Schuster–Visser no-go [2], whose null conclusion holds under the subsidiary conditions its authors state; the v_s^2 speed dependence is our calculation for this family, not a feature of the theorem.

3.2 Shift vorticity and algebraic type

The benchmark shows different algebraic types even at matched speeds and wall widths. To interpret this difference, we relate the Eulerian momentum to spatial derivatives of the shift. This identifies a sufficient condition for Type I and the additional stress information needed when momentum is nonzero.

For flat unit-lapse slices, the Eulerian momentum density obeys $8\pi|j| = \frac{1}{2}|\nabla \times (\nabla \times \beta)|$ (Lemma 1), where $\beta_i = g_{0i}$ is the ADM shift. Under the smoothness, entire-domain and bounded-vorticity hypotheses of Lemma 2, it vanishes everywhere exactly for gradients plus rigid rotation. This characterizes momentum, not algebraic type: the discriminant also involves ρ_n and S, and its type implication requires the splitting assumption of Section 2.2. On the slice we split the covariant gradient of the shift into expansion, shear, and vorticity parts, as a fluid velocity field is decomposed,

$$D_i\beta_j = \tfrac{1}{3}\,\theta_\beta\,\gamma_{ij} + \sigma^\beta_{ij} + \omega^\beta_{ij}, \qquad \omega^\beta_{ij} = D_{[i}\beta_{j]} = \partial_{[i}\beta_{j]}, \tag{29}$$

where the vorticity part is one half of the exterior derivative of the shift one-form, so it follows from first metric derivatives and transforms tensorially under time-independent spatial coordinate changes. The wall summaries use $\langle X\rangle_W = (\int_W X\, dV)/(\int_W dV)$ on $t = 0$, $W = \{0.1 \le f \le 0.9\}$,

and the dimensionless vorticity fraction

$$\mathcal{R}_\omega = \frac{\langle \omega_\beta^2 \rangle_W}{\langle \theta_\beta^2/3 + \sigma_\beta^2 + \omega_\beta^2 \rangle_W}.$$

Thus the wall integrals are formed before taking the ratio. Expansion and shear fractions use the same denominator. For a fixed comoving wall, a speed-linear shift and a speed-independent spatial metric make these fractions independent of nonzero v_s, provided the denominator is nonzero. Here $\sigma_\beta^2 = \sigma^\beta_{ij}\sigma_\beta^{ij}$ and $\omega_\beta^2 = \omega^\beta_{ij}\omega_\beta^{ij}$ denote full contractions.

Table 4(a) and Figure 2 report the wall-restricted, proper-volume-weighted decomposition. Rodal's analytic shift is irrotational by Lemma 3; the implemented regularization gives $\mathcal{R}_\omega = 0.000$. The sufficient condition for Type I is classical: an irrotational shift on a unit-lapse, flat slice gives $j = 0$ and a symmetric spatial stress block [2, 5]. The remaining drives have nonzero shift vorticity ($\mathcal{R}_\omega = 0.333$ Alcubierre, 0.440 Van den Broeck, 0.508 Natário) and all have sampled Type-IV wall regions. Natário has *zero* shift expansion (its computed expansion fraction is 0.000) and the largest vorticity fraction in this sample.

The association between $\mathcal{R}_\omega$ and type is empirical within this family. Lemma 2 concerns vanishing momentum: j depends on $\nabla \times \omega$, whereas the decoupled momentum discriminant also involves the energy density and parallel stress. The transverse channel of Corollary 1 requires the full spatial stress. Rigid rotation illustrates the distinction: it has $\mathcal{R}_\omega = 1$, $\nabla \times \omega = 0$ and $j = 0$, and describes Minkowski spacetime in rotating coordinates. The exact curvature coefficient in Section 3.6 likewise depends on derivatives of ω. The vorticity fraction alone supplies no criterion for type, energy-condition violation or curvature scaling.

The curl of the shift vorticity fixes j, while the Hamiltonian constraint fixes ρ_n from the symmetric shift gradient. The spatial stress also contains evolution and Lie-derivative terms. Where $\hat{\jmath}$ is a principal axis of S, the momentum plane decouples and $2|j| > |\rho_n + S_\parallel|$ gives a Type-IV pair. Without this splitting, $\Delta < 0$ still supplies an axial NEC-violation witness but does not determine the algebraic type. Table 4(b) compares the full eigensolver's imaginary part with the decoupled-plane prediction at a specified wall point for each drive.

Under the splitting hypothesis with $\Delta < 0$, the positive imaginary part is

$$\operatorname{Im}\lambda = \tfrac{1}{2}\sqrt{-\Delta} = \tfrac{1}{2}\sqrt{4|j|^2 - (\rho_n + S_\parallel)^2}\,, \tag{30}$$

which gives $\operatorname{Im}\lambda \approx |j|$ where the momentum dominates. On a flat unit-lapse slice the Eulerian momentum density is fixed by the shift through an exact identity.

Lemma 1 (Momentum density is the curl of the shift vorticity)**.** *Let $\beta_i = g_{0i}$ be a C^2 shift on a region with $\alpha = 1$ and a time-independent flat spatial metric $\gamma_{ij} = \delta_{ij}$. Write $h^a{}_b = \delta^a{}_b + n^a n_b$ for the spatial projector. The Eulerian momentum density $j^i = -h^i{}_a T^a{}_b n^b$ obeys, in magnitude and to all orders in β,*

$$8\pi\,|j| = \tfrac{1}{2}\,|\nabla \times (\nabla \times \beta)| = \tfrac{1}{2}\,|\nabla \times \omega|, \qquad \omega = \nabla \times \beta. \tag{31}$$

Consequently an irrotational shift ($\omega = 0$) and a spatially uniform rotation ($\nabla \times \omega = 0$) both have $j = 0$. The momentum is determined by $\nabla \times \omega$, rather than by the vorticity alone.

Proof. With $K_{ij} = -\frac{1}{2}\mathcal{L}_n\gamma_{ij}$ and the $+\beta$ convention, $K_{ij} = \frac{1}{2}(\partial_i\beta_j + \partial_j\beta_i)$ and $K = \nabla\cdot\beta$. The momentum constraint gives

$$8\pi j_i = \partial_j K_{ij} - \partial_i K = \tfrac{1}{2}\big[\nabla^2\beta - \nabla(\nabla\cdot\beta)\big]_i = -\tfrac{1}{2}\big[\nabla \times (\nabla \times \beta)\big]_i.$$

Taking the Euclidean norm proves the identity. □

Santiago–Schuster–Visser [2], equation (4.8), give this identity as $f_i = \frac{1}{16\pi}[\nabla \times (\nabla \times v)]_i$ in their flow convention. They also establish that an irrotational shift gives Type-I stress-energy. To identify all shifts with vanishing momentum, a local condition is not enough: the following converse uses the entire slice and bounded vorticity. It includes rigid rotation.

Lemma 2 (Shifts with vanishing momentum)**.** *For $\alpha = 1$ and a time-independent spatial metric $\gamma_{ij} = \delta_{ij}$, let the shift β be smooth on $\mathbb{R}^3$ with bounded vorticity $\omega = \nabla \times \beta$. Then*

$$j \equiv 0 \iff \beta = \nabla\phi + \Omega \times \mathbf{x} \quad \text{for a scalar } \phi \text{ and a constant vector } \Omega. \tag{32}$$

A stationary shift consisting of rigid rotation alone gives $K_{ij} = 0$ and a flat spacetime. If the vorticity tends to zero at infinity, then $\Omega = 0$.

Proof. If β has that form then $\omega = 2\Omega$ is constant, so $\nabla \times \omega = 0$ and $j \equiv 0$ by Lemma 1. Conversely $j \equiv 0$ gives $\nabla \times \omega = 0$, while $\nabla \cdot \omega = 0$ holds for any curl, so $\nabla^2 \omega = \nabla(\nabla \cdot \omega) - \nabla \times (\nabla \times \omega) = 0$: each Cartesian component of ω is harmonic on $\mathbb{R}^3$ and bounded, hence constant by Liouville's theorem. Writing $\omega = 2\Omega$ and using $\nabla \times (\Omega \times \mathbf{x}) = 2\Omega$ leaves $\nabla \times (\beta - \Omega \times \mathbf{x}) = 0$ on a simply connected domain, so that difference is a gradient. Finally $\partial_i (\Omega \times \mathbf{x})_j = \epsilon_{jki} \Omega_k$ is antisymmetric in (i, j), so $K_{ij} = \partial_{(i} \beta_{j)}$ vanishes on it. □

The controlled differential-rotation experiment uses $\beta = c(-y, x, 0) e^{-(x^2+y^2+z^2)/2}$ at $(x, y, z) = (0.5, 0.5, 0)$, with $c \in [0, 0.4]$. Fitting its imaginary eigenvalue against $\omega = (\omega_{ij}\omega^{ij})^{1/2} = |\nabla \times \beta|/\sqrt{2}$ gives the empirical slope $\kappa \approx 0.060$. This coefficient belongs to that profile and sample point. Under the momentum-plane splitting hypothesis, equation (30) instead gives the exact relation $\operatorname{Im} \lambda = |j| \sqrt{1 - (\rho_n + S_\parallel)^2/(4|j|^2)}$. For $|\rho_n + S_\parallel| \ll 2|j|$, its leading fractional difference from $|j|$ is $(\rho_n + S_\parallel)^2/(8|j|^2)$. An irrotational flat-slice unit-lapse shift has $j = 0$ and Type I stress-energy, as used in Rodal's construction [6].

Lemma 3 (Type I stress-energy for the irrotational profiles)**.** *Write the Rodal shift in the orthonormal spherical frame of (49), $\hat{\beta}_r = -v_s F(r_s) \cos\theta$, $\hat{\beta}_\theta = v_s G(r_s) \sin\theta$. Its only possibly nonvanishing curl component is the azimuthal one, and*

$$(\nabla \times \beta)_\phi = \frac{v_s \sin\theta}{r_s} \big[(r_s G)' - F \big]. \tag{33}$$

For $v_s \neq 0$, the shift is curl-free on each sphere exactly when $F = (r_s G)'$. The profiles (46) and (50) satisfy that relation identically in (r_s, σ, R_b). Hence $\nabla \times \beta \equiv 0$, so $j \equiv 0$ by Lemma 1, so $T^a{}_b$ is block diagonal $(-\rho_n) \oplus S_{ij}$ with S symmetric: its spectrum is real and the slice normal is a timelike eigenvector. The drive is Hawking–Ellis Type I at every point of every slice, at every v_s. The same holds for the Garattini–Zatrimaylov bubble on its matching condition, whose shift is a sum of two radial gradients (Appendix D) and therefore has $j = 0$ by Lemma 1.

Proof. With $\beta_\phi = 0$ and axial symmetry, the radial and polar curl components vanish termwise and the azimuthal one is $r_s^{-1}[\partial_{r_s}(r_s \hat{\beta}_\theta) - \partial_\theta \hat{\beta}_r]$, which is (33). Writing $\Delta_\sigma(r_s) = \ln\cosh\sigma(r_s - R_b) - \ln\cosh\sigma(r_s + R_b)$, equation (50) is $G = -\Delta_\sigma/(2 r_s \sigma \tanh\sigma R_b)$, so $(r_s G)' = -\Delta_\sigma'/(2\sigma\tanh\sigma R_b) = [\tanh\sigma(r_s + R_b) - \tanh\sigma(r_s - R_b)]/(2\tanh\sigma R_b) = f(r_s) = F$. The block-diagonal step is the $j = 0$ case of the momentum decomposition of Section 2.2. □

The conclusion for the irrotational drive is Rodal's own Proposition 2 [6], proved there under the same assumptions; the relation $F = (r_s G)'$ is the irrotationality condition his construction is built to satisfy, written in the conventions used here. In this form the label is a continuum property. The numerical Rodal implementation uses small-radius regularizations (Appendix D); its residual curl is a numerical effect; the exact zero applies to the analytic profile.
The harmonic-gradient implication of Barzegar, Buchert and Vigneron [18] applies under its R-Warp hypotheses, including the stated asymptotic assumptions. Rodal's potential is nonharmonic, so that implication gives no conclusion for this drive.

The sign of Δ determines the momentum-plane type under the splitting hypothesis. The reported Type-IV labels are taken from the full four-dimensional eigensolver, not from the sign of Δ. A separate unweighted count comparison uses $N = 60$, clustering $a = 1.2$, $t = 0$ and $v_s = 0.5$: bounds ± 3 with $R_b = 1$, $\sigma = 8$ for the first three drives, and bounds ± 300 with $R_b = 100$, $\sigma = 0.03$ for Rodal. The indicator $\Delta < 0$ agrees with the Type-IV label at 100% of Rodal wall points, 97.5% of Alcubierre's, 88.4% of Natário's and 76.2% of Van den Broeck's, with full-grid agreements 98.9%, 99.7% and 81.6% for Alcubierre, Natário and Van den Broeck, respectively. The sign $\Delta < 0$ supplies a NEC-violation witness without deciding Type IV. Numerical disagreement alone does not identify its cause or test the splitting hypothesis. Natário and Van den Broeck have 368 and 752 wall points labeled Type IV with $\Delta \geq 0$, respectively. The sampled Type-IV fraction varies with speed and wall geometry: only 7% of the Van den Broeck wall is Type-IV at $v_s = 0.1$, before its Type-I → Type-IV transition. Maeda [67] showed that in a static spacetime solving the Einstein equations the stress-energy is Type I wherever the hypersurface-orthogonal Killing field is timelike. Related restrictions hold on suitable Killing horizons and symmetry axes [17]. An established Type-IV tensor excludes staticity under those hypotheses; laboratory-coordinate time dependence alone would not establish this.

Table 4. *(a)* Wall-restricted, proper-volume-weighted decomposition of the ADM shift gradient (Eq. 29) into expansion, shear, and vorticity fractions, with the resulting wall Type-IV range over $v_s \in [0.1, 2.5]$. The vorticity fraction $\mathcal{R}_\omega$ is independent of v_s. *(b)* The momentum-density discriminant at the matched wall sample ($v_s = 0.5$, off-axis wall point). The ratio $2|j|/|\rho_n + S_\parallel| > 1$ is equivalent to $\Delta < 0$ and gives Type IV under the momentum-plane splitting hypothesis. The measured $\mathrm{Im}\,\lambda$ is set beside the momentum-plane prediction $\frac{1}{2}\sqrt{\max(0, -\Delta)}$ at this sample; the wall-maximum is in Table 12.

		Shift-gradient fraction			Wall Type IV
	Metric	Expansion	Shear	Vorticity $\mathcal{R}_\omega$	range (%)
(a)	Alcubierre	0.111	0.556	0.333	75–100
	Natário	0.000	0.492	0.508	56–99
	Van den Broeck	0.193	0.367	0.440	7–93
	Rodal	0.485	0.515	0.000	0–0

	Metric	Wall type	$2\|j\|/\|\rho_n + S_\parallel\|$	$\mathrm{Im}\,\lambda$	$\frac{1}{2}\sqrt{\max(0, -\Delta)}$
(b)	Rodal	I	0.00	0.000	0.000
	Van den Broeck	IV	0.89	0.058	0.000
	Alcubierre	IV	16.09	0.079	0.079
	Natário	IV	6.89	0.578	0.585

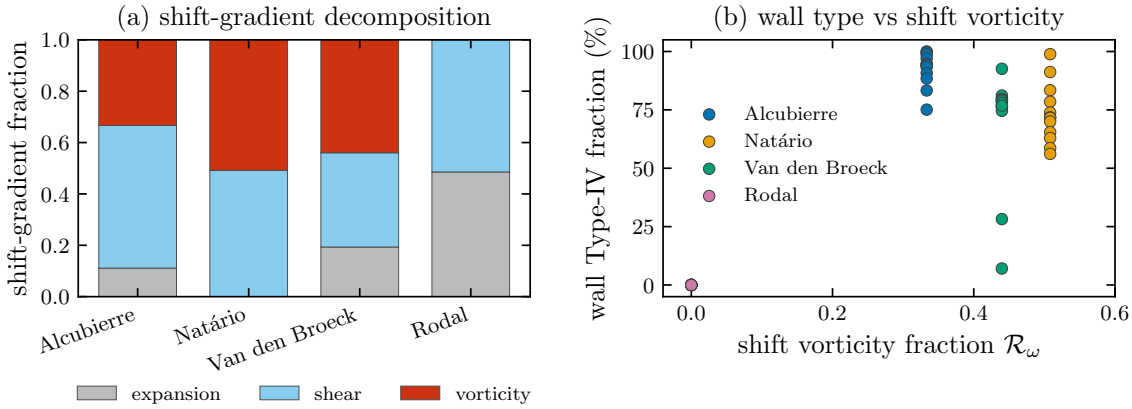

Figure 2. Shift-gradient decomposition and sampled type structure ($R_b = 1$, $\sigma = 8$, $t = 0$, $f \in [0.1, 0.9]$). (a) Fractions of the wall-integrated gradient norm ($N = 60$). Natário has zero expansion; Rodal's vorticity is below plot resolution, consistent with its ideal irrotational profile. (b) Proper-volume-weighted Type-IV fractions ($N = 100$) at the sampled speeds $v_s \in [0.1, 2.5]$, paired with each drive's speed-independent $\mathcal{R}_\omega$. Alcubierre has a smaller $\mathcal{R}_\omega$ but a larger Type-IV fraction than Natário. The vorticity fraction alone does not determine type.

3.3 Speed dependence of the momentum discriminant

The density and spatial stress scale quadratically with speed on the fixed flat-slice profiles, while the momentum scales linearly. Their competition therefore changes with speed. The discriminant expresses this change through the local threshold $v_\star$, whose coefficients can be computed at one speed.

At every wall point we compute $a = (\rho_n + S_\parallel)/v_s^2$ and $d = |j|/v_s$ on the wall-resolved benchmark grid, form the closing speed $v_\star = 2d/|a|$, and evaluate F_{mom} of (20) at the twelve speeds of the velocity sweep. One grid evaluation fixes the whole curve.

We compare the coefficients at $v_s = 0.5$ and $v_s = 1$; their relative changes appear in Table 5. Flat unit-lapse slices with the specified stationary or translating profile have the exact scaling of Section 2.2. Van den Broeck has a curved spatial metric, so its single-speed field is included only as a numerical comparison. For the ideal Rodal shift $j = 0$, the momentum channel is absent.

Table 5 compares F_{mom} with the sampled Type-IV fraction. Its signed gap is a difference of aggregate fractions. The calculation does not check principal-axis alignment or the pointwise implication $\Delta < 0 \Rightarrow$ Type IV at each speed. Consequently the gap does not establish a bound or a volume assigned to a transverse mechanism. The numerical closing field treats $|a| \le 10^{-12} \max_{\mathrm{wall}} |a|$ as zero; the exact criterion above distinguishes $a = 0$ from every nonzero value.

Table 5. The momentum channel as a speed-independent field. δa is the relative change in $a = (\rho_n + S_\parallel)/v_s^2$ between $v_s = 0.5$ and $v_s = 1$, a check of the scaling assumption. $\tilde{v}_\star$ is the median finite closing speed over the wall. $F_{\rm mom}$ (%) is (20), computed once at $v_s = 1$ and read at each speed; "gap" is the measured wall Type-IV fraction minus $F_{\rm mom}$, in percentage points. Neither the sign of this gap nor coefficient agreement verifies the splitting hypothesis.

			$v_s = 0.1$		0.5		1.0		2.5	
Metric	δa	$\tilde{v}_\star$	$F_{\rm mom}$	gap	$F_{\rm mom}$	gap	$F_{\rm mom}$	gap	$F_{\rm mom}$	gap
Alcubierre	2e-15	2.48	99.9	+0.1	95.5	+3.3	84.6	+9.6	48.8	+26.3
Natário	1e-14	1.24	98.4	+0.4	73.8	+9.6	57.8	+13.7	31.5	+24.7
Van den Broeck	2e+00	1.74	100.0	-92.9	99.0	-17.9	90.8	-11.9	35.2	+41.6
Rodal	0e+00	0.00	0.0	+0.0	0.0	+0.0	0.0	+0.0	0.0	+0.0

Table 6. Energy condition analysis at $v_s = 0.5$ on a 50^3 grid ($\zeta_{\max} = 5$ throughout). *Total*: percentage of grid points violating each condition on the robust margin of Section 2.2. *Missed*: percentage of grid points where the Eulerian analysis reports no violation the robust check finds, so the Eulerian violated fraction is the total minus the missed; 0.0% means all violations are detected to the displayed precision, not that there are none. Bold marks missed fractions $\geq 1\%$.

	Total violated (%)				Missed by Eulerian (%)			
Metric	NEC	WEC	SEC	DEC	NEC	WEC	SEC	DEC
Schwarzschild	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Alcubierre	6.2	9.4	9.4	9.6	0.0	0.0	**7.2**	0.0
Van Den Broeck	6.6	7.8	7.9	8.1	0.1	0.4	**1.8**	0.3
Natário	7.7	12.0	12.0	12.2	0.0	0.0	**3.2**	0.0
Rodal	87.0	87.0	87.0	99.9	**1.6**	**15.6**	**28.0**	**28.5**

3.4 Single-frame and finite-observer diagnostics

The benchmark tests the energy conditions over all observers. We now ask how many of these violations a single observer or a finite observer sample misses, using the same stress-energy calculation for both tests.

Two examples show the size of the effect. For Alcubierre at $v_s = 0.5$ the Eulerian and all-observer NEC violation sets coincide on the sampled native grid at the stated detection threshold (the sampled all-wall NEC minimum is -0.628 over null directions and -0.625 for the six Eulerian directions on the native uniform grid, distinct from the Type-I invariant deficit $\min(\rho + p_i) = -0.158$ of Table 2). The WEC contractions differ much more: the minimum found at $\zeta_{\max} = 5$ is ~ -3460, compared with -0.038 for the Eulerian observer. As Section 2.5 explains, this magnitude depends on the rapidity cap. Van den Broeck, whose conformal factor acts on the spatial metric, is the only drive with sampled Type-IV wall regions with missed violations across all four conditions on the uniform-grid diagnostic (Table 6); its SEC has the largest conditional miss among its conditions ($f_{\rm miss|viol} = 22.6\%$; Figure 3), and for Alcubierre the SEC is the only missed condition at all.

Figure 4 compares a finite observer sample with the Type-I algebraic criterion within the same implementation. It isolates the need for an all-observer test without requiring an observer optimization for the reference set.

We define the *missed violation fraction* for a given energy condition as

$$f_{\rm miss} = \frac{\#\{x \in \text{grid} : m_{\rm Euler}(x) \geq 0 \ \wedge \ m_{\rm rob}(x) < 0\}}{\#\{x \in \text{grid}\}}, \tag{34}$$

i.e. the fraction of grid points the Eulerian frame reports as satisfied ($m_{\rm Euler} \geq 0$) but the observer-independent verdict marks violated (the exact algebraic slack at Type I, and at Type-IV points the null violation the label implies; Section 2.2). The unconditional $f_{\rm miss}$ (Table 6) is a *domain-restricted diagnostic*: it scales with each metric's wall thickness and grid volume and is *not* a cross-metric severity measure. The appropriate cross-metric statistic is the wall-restricted conditional miss rate $f_{\rm miss|viol}$, the fraction of robust-violated points the Eulerian analysis misses, (missed)/(total violated), which does not dilute the failure rate over the satisfied points. It is reported on the matched benchmark (Table 7(a)) and in Section 3.1; the matched and invariant computations, on the same wall-resolved graded family, agree to within grid resolution (Rodal wall DEC miss 73.6–74.0%). Table 6 gives the within-metric Eulerian-vs-robust comparison; note SEC misses are independent of the others (the combination $\rho + \sum_i p_i$ can be negative while each $\rho + p_i \geq 0$).

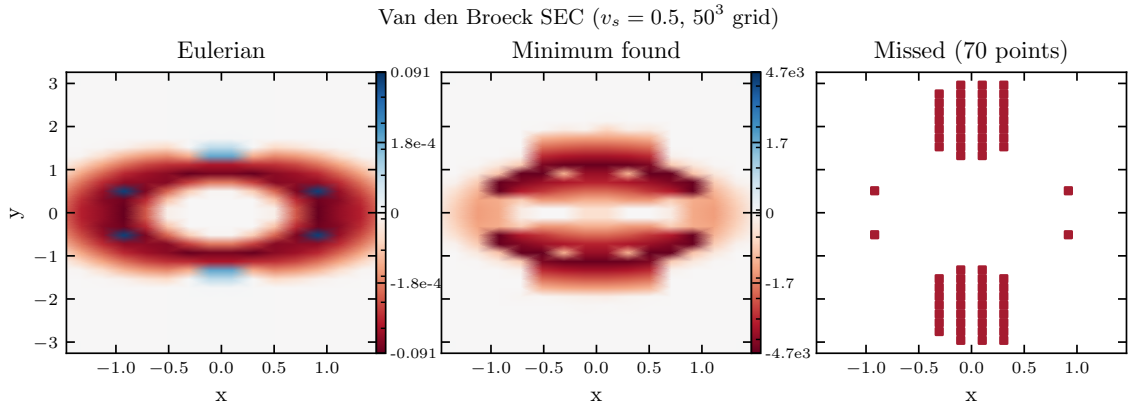

Figure 3. Van den Broeck SEC at $v_s = 0.5$ on the 50^3 native grid. Left: Eulerian contraction. Center: observer minimum found with $\zeta_{\max} = 5$. Their symmetric logarithmic color scales differ; the capped magnitude is not an invariant severity. Right: all 70 missed sample points in the plane $z = 5/49 \approx 0.102 R_b$, marked in red without interpolation. At these points the Eulerian contraction is nonnegative and the boosted contraction is below -10^{-10}. The full-grid miss is 1.8%, or 22.6% of detected violations (Table 6); these percentages are not slice counts.

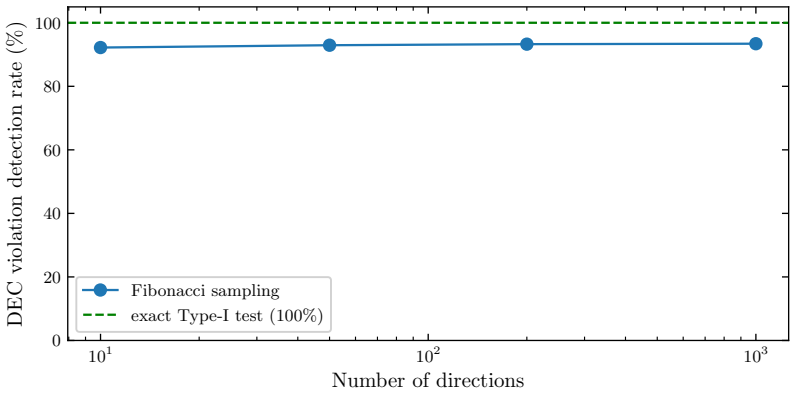

Figure 4. Finite-observer DEC detection for Rodal ($R_b = 100$, $\sigma = 0.03$, $v_s = 0.5$; 25^3 grid). Each Fibonacci direction uses ten rapidities in $[0, 5]$. Detection rate (circles) reaches 93.4% at 10^4 observer samples per point. The dashed 100% reference is the Type-I criterion on the 15 624 violating points of the 15 625 sampled; both diagnostics use threshold -10^{-10}. The residual miss combines angular and rapidity sampling, the cap, and threshold effects. It is not an isolated angular-sampling error.

Table 7. Matched-parameter cross-metric benchmark. *(a)* All retained metrics at common family parameters ($R_b = 1$, $\sigma = 8$) on identical compact bounds, at the finest resolution of the ladder of part (b). The reported quantity is the volume-weighted *wall-restricted* conditional miss rate: Eulerian-satisfied but robust-violated points, as a fraction of robust-violated points within the wall $f \in [0.1, 0.9]$. *(b)* Per-metric resolution validation of that benchmark: the wall-restricted DEC miss rate at three levels of the wall-resolved graded ladder ($N = 80, 100, 120$, giving 5.9, 7.6 and 8.9 cells across the 10–90% wall) and the maximum absolute deviation from the three-grid mean, in percentage points. With $\bar{f}$ the mean in percent, "Stable" means this deviation is at most $\max(0.5, 0.05|\bar{f}|)$ percentage points. The miss rate is a non-smooth thresholded fraction, so it has a stability spread and not a Richardson order.

		Wall	Missed by Eulerian, wall-restricted (%)			
	Metric	pts	NEC	WEC	SEC	DEC
(a)	Alcubierre	22688	0.0	0.0	18.7	0.0
	Natário	22688	0.0	0.0	0.0	0.0
	Van den Broeck	22688	16.7	57.4	8.6	53.1
	Rodal	22688	12.5	72.2	19.5	73.6

		Wall DEC miss (%) at $N =$			Max dev	Stable
	Metric	80	100	120	(pp)	
(b)	Alcubierre	0.0	0.0	0.0	0.00	✓
	Natário	0.0	0.0	0.0	0.00	✓
	Van den Broeck	56.1	51.3	53.1	2.58	✓
	Rodal	75.7	74.0	73.6	1.27	✓

Velocity dependence of the Eulerian comparison. Section 3.1 reports the sampled type structure across the luminal transition. In the subluminal Alcubierre observer comparison, the Eulerian frame detects all sampled NEC/WEC violations to the displayed precision (missed $= 0.0\%$). The capped contractions become more negative as speed increases.

For Rodal at its native parameters ($R_b = 100$, $\sigma = 0.03$) the Eulerian frame misses $\sim$28.5% of DEC and $\sim$15.6% of WEC violations as a fraction of the full 50^3 grid (Table 13(c)). This unconditional grid fraction is not the wall-restricted conditional miss fraction of the matched-parameter benchmark ($\approx$74% DEC, $\approx$73% WEC; Section 3.1), which conditions on the violating points within the active wall. The discrepancy is systematic across the velocity range and strongest for DEC and WEC.

Finite observer sampling and the algebraic criterion. WarpFactory [13, 41, 42] takes the minimum energy-condition value over a finite sample of observer directions and velocities at each grid point. The retained Fibonacci-lattice comparison uses Rodal at $R_b = 100$, $\sigma = 0.03$ and $v_s = 0.5$, with ten rapidities between zero and five. At 10^3 directions per point it detects 93.4% of the 15 624 points whose Type-I algebraic DEC slack is below -10^{-10} (Figure 4); the minimum algebraic slack is -2.0446×10^{-6}. The sampled diagnostic also uses threshold -10^{-10}. This measures the shortfall of that finite observer search; the calculation does not separate the effects of angular sampling, rapidity sampling, the cap and numerical thresholds. The algebraic criterion supplies the reference set without an observer search.

3.5 Averaged and quantum conditions

The pointwise classical results do not determine averaged conditions or semiclassical bounds. We report finite-segment null-energy integrals and a Ford–Roman reference estimate along a coordinate-static wall worldline (Table 8, Figure 5).

Finite-segment null-energy diagnostics. Table 8 reports two finite-window calculations at $R_b = 1$, $\sigma = 8$, $v_s = 0.5$. In the coordinate-path diagnostic we sample $(t, x, y, z) = (\lambda, -8 + \lambda, b, 0)$ and contract T with a null-projected direction at each node. That projected vector need not be the path tangent. This calculation has no complete-geodesic ANEC interpretation. Its parameter interval is $[0, 43.94822954822955]$ in units of R_b, with 80 initial impact parameters in $[10^{-3}, 2.5]$ and the selected-path refinement described below.

For the geodesic diagnostic we integrate the canonical Hamiltonian $H = \frac{1}{2} g^{ab} p_a p_b$ with Tao's extended-phase-space method and a fourth-order Yoshida composition [55, 56]. The initial position is $(0, -8R_b, b, 0)$ with initial spatial direction $+\hat{x}$. We fix the affine scale separately for every ray by

$$-g_{ab} k^a n^b = 1 \tag{35}$$

at its initial point. Under a constant rescaling $k \mapsto ck$, $c > 0$, with the same geometric endpoints, $\int T(k,k)\, \mathrm{d}\lambda$ scales by c. Its magnitude therefore depends on this normalization. Let $s > 0$ be the initial spatial scale factor that enforces this normalization. The affine bounds are $[0, L/s]$, with $L = 43.5849609375$ for Alcubierre, Natário, and Rodal, and $L = 47.5927734375$ for Van den Broeck. For Alcubierre, Natário, and Van den Broeck, $s = 1$; for Rodal it is 1.0000000009460461 on the near-axis ray and 1.0054270723616685 on the minimum-found ray. All lengths are in units of R_b.

Each metric's integration window is selected from a probe ray, with a factor-two margin after its outward crossing of $r_s = 3R_b$. This is a window-selection rule only. The shape-function bound outside that radius does not bound the stress contraction. In particular, Rodal's angular profile has $G(r_s) \sim R_b/[r_s \tanh(\sigma R_b)]$, so its shift and stress have algebraic tails despite the radial tanh profile. Neither omitted affine tail is bounded here. The reported integral is $I_{\mathrm{seg}} = \int_{\lambda_-}^{\lambda_+} T_{ab} k^a k^b \, \mathrm{d}\lambda$, not an integral over a complete geodesic.

The impact-parameter search starts with 50 equally spaced values on $[10^{-3}, 5]$. We exclude incomplete rays and rays failing the relative null check below, refine a bracket of finite candidates locally, and compare that result with the lowest finite coarse candidate. The lower step-stable value is reported as a *minimum found*. For Natário, 9 of the 50 fan rays are excluded. Its lowest retained candidate, at $b = 1.429286$, is unbracketed; the integral is step-stable but the impact-parameter search has no convergence claim. This search establishes neither a global minimum nor geodesic completeness.

At fixed impact parameter and affine endpoints, selected rays are recomputed starting at step densities 16384, 32768 and 65536 per reference span $16/s$, where s is the initial spatial scale. Further doublings are used when needed. Table 13(a) gives the densities and observed changes for

Table 8. Averaged and quantum diagnostics at matched parameters ($R_b = 1$, $\sigma = 8$, $v_s = 0.5$). *(a)* Finite-segment null-geodesic integrals, with the symplectic integrator at the finest step densities reported in Table 13(a), with quadrature at every step. $\int T_{ab}k^a k^b \, d\lambda$ is given on $b = 10^{-3}$ and at the minimum found (with its b^*); Natário's selected candidate is unbracketed. The maxima cover retained finite fan rays and selected refinements; $\max|g(k,k)|$ is the absolute off-cone deviation and $\max|\Delta E_K/E_K|$ the worst relative drift of the conserved Killing energy. *(b)* The coordinate-path integral with projected null directions $\int T_{ab}k^a k^b \, d\lambda$ at $b = 10^{-3}$ and at the minimum found over its coordinate-path scan, with selected-ray quadrature refinement, with the Minkowski ray = 0 to $<10^{-8}$, beside the Ford–Roman columns: the most-negative static-observer wall energy density $R_b^2\rho_{\min}$ and the short-window coefficient c_{metric} in equation (36), reported for the selected Alcubierre and Rodal worldlines (a dash elsewhere).

(a)

Metric	$b = 0.001$	minimum found (b^*)	$\max\|g(k,k)\|$	$\max\|\Delta E_K/E_K\|$	method
Alcubierre	-0.0772	-0.1462 (0.41)	$1.5e-10$	$1.0e-10$	symplectic
Natário	-0.1797	-0.5404 (1.43)	$1.1e-06$	$6.7e-07$	symplectic
Van den Broeck	-0.0536	-0.0626 (0.60)	$3.9e-11$	$4.2e-11$	symplectic
Rodal	$+0.0424$	-0.0017 (2.57)	$1.4e-10$	$9.5e-11$	symplectic

(b)

	Finite-segment $\int T_{ab}k^a k^b \, d\lambda$		Short-window estimate	
Metric	$b = 0.001$	min found (b^*)	$R_b^2\rho_{\text{static,min}}$	c_{metric}
Alcubierre	-0.161	-0.793 (0.86)	-0.0876	0.57
Natário	-0.588	-3.51 (0.82)	–	–
Van den Broeck	$+0.0217$	-0.0987 (0.82)	–	–
Rodal	$+0.0439$	-0.0136 (1.27)	-0.00704	1.08

the minimum-found rays; the finest value is reported. The near-axis rays $b = 10^{-3}$ use all three initial densities. Their finest-pair absolute changes are 7.90×10^{-15}, 2.07×10^{-13}, 1.26×10^{-13} and 2.18×10^{-13} for Alcubierre, Natário, Van den Broeck and Rodal, respectively. Selected coordinate-path values use sample densities 1024, 2048 and 4096 per reference span 16.

The null check requires $\max|g(k,k)| < 10^{-6}D$, where $D = \max_{\text{nodes}}[(\max_{ab}|g_{ab}|)(\max_a|k^a|)^2]$; a zero D is replaced by one. Thus the test is relative, whereas the table prints the absolute residual. A second monitor is the relative drift of $E_K = -g(k,K)$, where $K = \partial_t + v_s\partial_x$ is Killing for the translating benchmark metrics. Both use every quadrature node. The displayed maxima cover retained finite fan rays and selected refinements. They are numerical diagnostics, not trajectory or quadrature error bounds.

Quantum inequality. At a fixed wall point the passing bubble presents a coordinate-static observer with a temporary negative-energy pulse. As a reference estimate, we use the four-dimensional Minkowski inertial bound for a free massless minimally coupled scalar field, with Lorentzian sampling, $\int \rho(\tau)\, f(\tau)\, d\tau \geq -C\hbar/\tau_0^4$, with $C = 3/(32\pi^2)$ and $f(\tau) = \tau_0/[\pi(\tau^2 + \tau_0^2)]$ the normalized sampling function. In geometric units $G = c = 1$, $\hbar = \ell_P^2$; retaining it distinguishes the Planck length from the bubble scale. Balancing the wall pulse against the bound gives a threshold set by the geometric mean of the two scales,

$$\tau_0^{\text{th}} \;\simeq\; c_{\text{metric}}\sqrt{\ell_P R_b}, \tag{36}$$

where $c_{\text{metric}} = (C/|R_b^2\rho_{\min}|)^{1/4}$ in the short-window approximation $\rho(\tau) \simeq \rho_{\min}$. Here $\rho = T_{00}/(-g_{00})$ is the density measured by $u = (-g_{00})^{-1/2}\partial_t$, defined where $g_{00} < 0$; it is generally different from the Eulerian density ρ_n. The plotted curves set $R_b = G = c = \hbar = 1$ and show the dimensionless diagnostic

$$\mathcal{M}(\tau_0) = R_b^2\left[\int_{\tau_-}^{\tau_+} \rho(\tau) f(\tau)\, d\tau + \frac{C\hbar}{\tau_0^4}\right].$$

Proper time is accumulated along the coordinate-static worldline, with $d\tau = \sqrt{-g_{00}}\, dt$; the numerical window extends approximately $10\tau_0$ on either side of its center. The Lorentzian kernel is not renormalized on this finite window; we do not bound the omitted stress tails. These Alcubierre and Rodal pulse diagnostics sample only selected worldlines. Their zero crossings cannot be rescaled to macroscopic R_b by equation (36). Curvature and acceleration corrections would be required for a quantum-energy bound on these worldlines [21, 63]. Neither diagnostic establishes complete-geodesic ANEC violation or a curved-spacetime quantum-energy bound.

3.6 Kinematically standardized constructions and a slice diagnostic

To compare constructions with different length scales, Table 9 applies the same pointwise energy-condition diagnostics to the regularized Fuchs shell [7] and the Garattini–Zatrimaylov de Sitter bubble [10], beside Alcubierre and Rodal. This panel standardizes kinematics only.

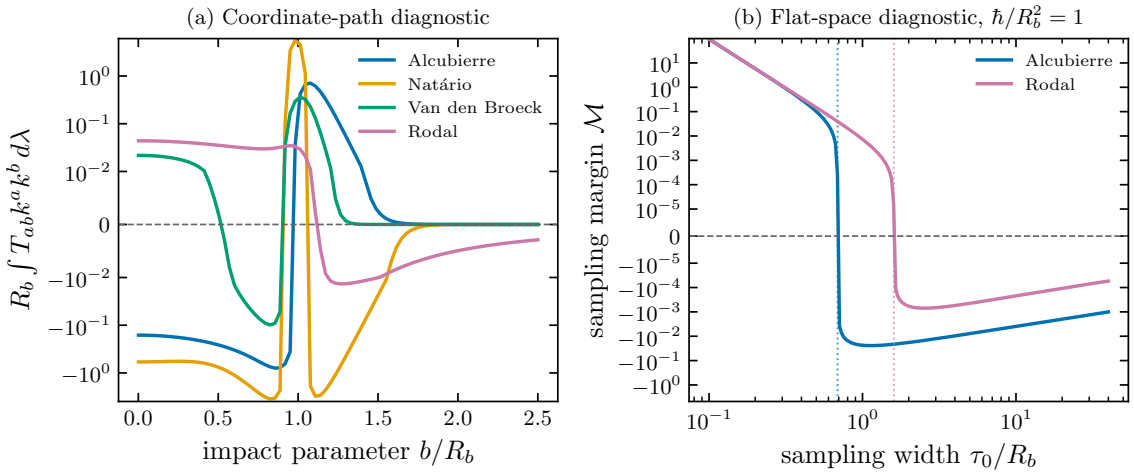

Figure 5. Finite-window diagnostics at $R_b = 1$, $\sigma = 8$, $v_s = 0.5$. (a) Coordinate-path integrals of the projected-null contraction versus impact parameter b; the projected vector need not be the path tangent. The strong dependence on b motivates the separate geodesic calculations in Table 8(a). (b) Sampling margin $\mathcal{M}(\tau_0)$ defined in the text for selected coordinate-static worldlines, with $\hbar = G = c = 1$. Dotted lines mark its zero crossings; both ordinates use symmetric logarithmic scales. Neither panel establishes complete-geodesic ANEC violation or a curved-spacetime quantum bound.

The standardized block uses $v_s = 0.02$, characteristic radius $R_c = 15\,\mathrm{m}$, and 10–90% profile width $W = 4.419943\,\mathrm{m}$, chosen from the Fuchs sigmoid. For the tanh profiles this gives $\sigma = 0.497115373\,\mathrm{m}^{-1}$. All rows use $t = 0$, the band $0.1 \le f \le 0.9$ and $r_{\max}/R_c = 2$. Each has its own axisymmetric grid ladder $(N_r, N_\mu) = (32, 32), (48, 48), (64, 64)$, clustered around its bubble center. The Garattini–Zatrimaylov center is $r_0 = v_s/H$, not the coordinate origin. The reference block uses the parameter sets specified in Appendix D; only the Fuchs numerical parameters are taken directly from its source.

Background curvature, shell compactness and matter content differ across the rows. The printed $R_c^2 \min(\rho + p_i)$ values are numerical diagnostics of the specified metrics and samples; the panel supplies no comparison of physical matter content or stress severity. For Garattini–Zatrimaylov, the matched-motion shift is a sum of radial gradients, so Lemma 3 gives exactly 100.0% Type I and 0.0% Type II. The repeated real eigenvalue does not produce a Jordan block. Its tabulated fractions use this exact result. The Fuchs row refers to the covariant modification and C^2 regularization specified in the appendix, at fixed smoothing widths.

Composite slice diagnostic. The composite in Table 10 combines three diagnostics on $t = 0$, at $R_b = 1$, $\sigma = 8$, $v_s = 0.5$: the minimum Type-I NEC algebraic slack on the $N = 100$ sweep grid, the proper-volume Type-IV fraction on the wall $0.1 \le f \le 0.9$, and the finite-segment basin-local minimum found above. With Alcubierre as reference, its sub-scores are

$$s_{\mathrm{NEC}} = \min\left(1, \frac{|\min(\rho + p_i)|}{|\min(\rho + p_i)|_{\mathrm{Alc}}}\right), \quad s_{\mathrm{IV}} = \phi_{\mathrm{IV}}, \quad s_{\mathrm{seg}} = \min\left(1, \frac{|I_{\mathrm{seg,min}}|}{|I_{\mathrm{seg,min}}|_{\mathrm{Alc}}}\right), \tag{37}$$

and

$$\mathcal{E} = [\max(s_{\mathrm{NEC}}, \epsilon_0) \max(s_{\mathrm{IV}}, \epsilon_0) \max(s_{\mathrm{seg}}, \epsilon_0)]^{1/3}, \qquad \epsilon_0 = 10^{-4}. \tag{38}$$

The affine normalization is equation (35); the initial surface, endpoints and search are specified in Section 3.5. No rapidity cap enters these three axes; the separate timelike observer plots use $\zeta_{\max} = 5$. The composite depends on the slice, mask, normalization, ratio caps and floor. In particular, when $\phi_{\mathrm{IV}} = 0$, its value scales as $\epsilon_0^{1/3}$ if the other sub-scores exceed the floor. It has no observer-independent severity interpretation. The individual sub-scores are reported so that the effects of these choices remain visible.

Speed scaling of the wall NEC deficit. The speed sweep suggests simple power laws, but a fitted exponent need not follow from the field equations. We first state the exact scaling under fixed geometric assumptions, then compare it with the numerical fits. For a flat unit-lapse translating profile $\beta = v_s F(\mathbf{x} - v_s t \mathbf{e}_x)$, the Eulerian density and spatial stress scale as v_s^2 at fixed comoving points while momentum scales as v_s (Section 2.2); when $j = 0$, the invariant deficit is

$$\min(\rho + p_i) = -C\, v_s^2\,, \tag{39}$$

with speed-independent C, positive where the wall infimum is negative. The wall domain must be fixed in comoving coordinates. Under these fixed-profile and $j = 0$ hypotheses, the tensor is Type I and every $(\rho + p_i)/v_s^2$ is speed-independent for $v_s \neq 0$. Zero momentum alone does not imply $C > 0$. For the full three-dimensional Rodal profile, $C \approx 0.768$ is computed numerically from that geometry. The numerical coefficient depends on the spatial resolution: $C = 0.768$ on the velocity sweep (Table 11(b)) against $C = 0.759$ on the finest rung of the wall-resolved ladder, where $\min(\rho + p_i) = -0.1898$ at $v_s = \frac{1}{2}$ (Table 12), an observed 1.2% spread. This spread is not an uncertainty bound.

For Alcubierre and Natário, j need not vanish and the Type-I subset itself can change with speed. Their fixed-exponent rows in Table 11(b) are empirical fits to sampled Type-I wall slacks. The deviations can reflect both momentum-dependent eigenvalues and sampling of a changing subset; the present data do not separate these effects. Van den Broeck's conformally curved slice lies outside the flat-slice hypothesis. For the flat-slice family at $v_s > 0$, the axial null contraction is $\rho_n + S_\parallel - 2|j| = a\, v_s^2 - 2d\, v_s$. Its speed-linear coefficient is set by the double curl of the shift profile and vanishes when $j = 0$. For the Natário geometry, the nonzero momentum follows from localization. Its shift is divergence-free ($\nabla\cdot\beta = 0$, zero expansion), so Lemma 1 reduces to $8\pi j = \frac{1}{2}\nabla^2\beta$ and the momentum density vanishes identically only if the shift is harmonic; a harmonic field tending to a constant at infinity equals that constant, which is Minkowski in translated coordinates. A divergence-free drive that approaches a uniform shift at infinity without being uniform therefore has $|j| \neq 0$ somewhere in space, and there the null-witness deficit acquires a nonzero negative linear coefficient $-2d = -2|j|/v_s < 0$. Vorticity alone would not have sufficed: a uniform rotation is vortical with $j \equiv 0$. For a gradient shift, $j \equiv 0$ by Lemma 1, with the harmonic sublocus Minkowski under the full R-Warp assumptions of Barzegar–Buchert–Vigneron. Equation (39) is pointwise. The integrated identity of Santiago–Schuster–Visser [2] applies on a flat unit-lapse slice $\mathbb{R}^3$, with integrable squared shift gradients and boundary terms vanishing at infinity. Integrating the Hamiltonian constraint $16\pi\rho_n = K^2 - K_{ij}K^{ij}$ by parts gives

$$\int \rho_n \, dV = -\frac{1}{16\pi} \int \omega_{ij}\omega^{ij} \, dV \leq 0, \qquad \omega_{ij} = \partial_{[i}\beta_{j]}, \tag{40}$$

with ω_{ij} the shift vorticity ($|\nabla \times \beta|^2 = 2\omega_{ij}\omega^{ij}$). A nonzero shift vorticity makes the right side strictly negative, so $\rho_n < 0$ on a set of positive measure and the weak energy condition fails. This integrated identity alone does not prove NEC violation. Vorticity is sufficient but not necessary for a violation: the ideal irrotational Rodal drive has $\int \rho_n \, dV = 0$. Its $\beta = O(r_s^{-1})$ and $\partial\beta = O(r_s^{-2})$ tails make the squared gradients integrable and the integration-by-parts boundary term vanish. It nevertheless has pointwise weak- and null-energy violations, with a negative wall deficit $\min(\rho + p_i) = -C\, v_s^2$ of equation (39) at nonzero speed. The integrated negative Eulerian energy also scales exactly as v_s^2, without the additional $j = 0$ assumption (Lemma 4).

Lemma 4 (Exact integrated negative-energy scaling)**.** *For a drive with $\alpha = 1$, a time-independent spatial metric $\gamma_{ij} = \delta_{ij}$, and shift $\beta^i = v_s\, F^i(\mathbf{x})$ (F independent of v_s on the slice) the Eulerian integrated negative energy $E_-(v_s) = \int_{\rho_n<0} |\rho_n| \, dV$ (with $\rho_n = T_{ab}n^a n^b$), on a fixed integration domain, with $E_-(1) < \infty$, obeys $E_-(v_s) = v_s^2\, E_-(1)$ identically. For a translating profile the domain and spatial coordinates are held fixed relative to the bubble.*

Proof. The extrinsic curvature $K_{ij} = \partial_{(i}\beta_{j)}$ is linear in v_s, so on the flat slice the Hamiltonian constraint $16\pi\rho_n = K^2 - K_{ij}K^{ij}$ gives $\rho_n = v_s^2\, \hat{\rho}(\mathbf{x})$ with $\hat{\rho}$ independent of v_s (Section 2.2). The proper volume element is v_s-independent there. For $v_s \neq 0$ the negative set is fixed and the v_s^2 prefactor comes out of the integral. At $v_s = 0$ the negative set is empty and both sides vanish. □

The free-exponent fit of this integral returns $p = 2.00$, $R^2 = 1$ for Alcubierre, Natário, and Rodal (Table 11(c)), the integrated companion to the pointwise deficit and the same slice-integrated measure Rodal reports [6]. Van den Broeck is the exception: its conformally curved slice violates the flat-slice hypothesis, so its negative energy has a v_s-independent conformal offset present even at rest. These speed laws follow from the scaling of the shift and the ADM constraints. An exact quadratic dependence also appears in the profile-specific integrated energy of Jusufi and Lobo [69]; it does not remove the classical energy-condition violations of that effective metric.

Curvature scaling. In this paragraph, $K = R_{abcd}R^{abcd}$ denotes the Kretschmann scalar, distinct from the trace of the extrinsic curvature. We write $C^2 = C_{abcd}C^{abcd} = K - 2R_{ab}R^{ab} + R^2/3$ for the squared Weyl tensor in four spacetime dimensions. Table 11(d) fits curvature magnitudes to $X_{\max} = A v_s^q$ over the sampled subluminal branch (Figure 6). Here $X_{\max}$ is a local maximum found

by unconstrained polishing from a wall-grid seed; it is not an enclosed maximum over the wall. These are finite-speed fits: the selected point can move, and subleading terms can affect the fitted exponent. Previous studies examined the individual Alcubierre and Natário curvature invariants [27, 28]; the matched panel uses a common profile scale and wall mask for the seeds. Rodal's fitted curves are quartic to the reported precision, while Alcubierre and Natário are close to quadratic. Van den Broeck already has curvature at rest and admits no comparable single power law over this range.

For the fixed flat-slice, unit-lapse profiles considered above, the leading coefficients follow from the speed scalings of the stress components. The trace-reversed Einstein equation gives $R_{ab}R^{ab} = 64\pi^2\, T_{ab}T^{ab}$ with $T_{ab}T^{ab} = \rho_n^2 - 2|j|^2 + |S|^2$ in the Eulerian frame, so the contribution of order v_s^2 to $R_{ab}R^{ab}$ is $-128\pi^2|j|^2$ (since $|j| \propto v_s$) and vanishes exactly at $j = 0$, in particular for an irrotational shift, leaving the v_s^4 growth of the energy-stress terms. This contraction is not sign-definite in Lorentzian signature (its leading vortical term is negative), so the fitted quantity is the locally maximized magnitude $|R_{ab}R^{ab}|$. The Weyl-squared and Kretschmann exponents involve the full Riemann tensor and are *not* fixed by that identity. One direction of the mechanism is elementary: for a fixed smooth stationary or rigidly translating profile on flat unit-lapse slices, the order-v_s part of the metric is the shift covector $h_{0i} = \beta_i$, and for an irrotational shift $\beta_i = v_s\, \partial_i \Phi$ this is pure gauge (the time relabeling $t \to t - v_s\Phi$), so the order-v_s linearized Riemann tensor vanishes and every quadratic invariant starts at v_s^4 or higher. Nonzero shift vorticity rules out this scalar time relabeling, but does not by itself imply a nonzero curvature coefficient. Rigid rotation, for example, is removed by a time-dependent spatial rotation. The Kretschmann and Weyl-squared coefficients have different kernels.

For the following coefficients write $\beta = v_s F$ and $\bar{\omega}_{jk} = \partial_{[j}F_{k]}$, so $\omega_{jk} = v_s\bar{\omega}_{jk}$. The notation $[X]^{(2)}$ denotes the coefficient of v_s^2. The only order-v_s Riemann components that survive are $R^{(1)}_{0ijk} = \partial_i\bar{\omega}_{jk}$, and contracting them gives

$$\big[K\big]^{(2)} = -4\sum_{ijk}\big(\partial_i\bar{\omega}_{jk}\big)^2, \qquad \big[C^2\big]^{(2)} = \big[K\big]^{(2)} + 4\sum_{i}\Big(\partial^j\bar{\omega}_{ji}\Big)^2. \tag{41}$$

The Kretschmann coefficient is *minus a sum of squares*. On the eight-dimensional space of admissible $\partial\bar{\omega}$ (nine components of $\partial_i\bar{\omega}_{jk}$ less the one closedness relation $\partial_{[i}\bar{\omega}_{jk]} = 0$) it has signature $(0, 8, 0)$: negative definite. No cancellation is available, and the converse therefore *holds* for the Kretschmann scalar: $\partial\bar{\omega} \neq 0$ forces $[K]^{(2)} \neq 0$. The Weyl-squared coefficient has signature $(0, 5, 3)$, negative semidefinite with a three-dimensional kernel, so cancellation *is* available and the converse fails. The shift $F = (0, 0, -x^2 - y^2)$ realizes it: $\partial\bar{\omega} \neq 0$ with $[K]^{(2)} = -16$ and $[C^2]^{(2)} = 0$.

The kernel has a direct tensor description. Write $\bar{\omega}_{jk} = \epsilon_{jkl}v_l$ with $v = \frac{1}{2}\nabla \times F$ and $M_{il} = \partial_i v_l$; the closedness relation is exactly $\operatorname{tr} M = 0$, so the admissible space is the eight-dimensional traceless matrices. Here $\|M\|^2 = \sum_{il} M_{il}^2$ and $\operatorname{sym} M = (M + M^{\mathsf{T}})/2$. In those variables (41) reduces to

$$\big[K\big]^{(2)} = -8\,\|M\|^2, \qquad \big[C^2\big]^{(2)} = -8\,\|\operatorname{sym} M\|^2. \tag{42}$$

Writing $(\operatorname{skew} M)_{il} = \epsilon_{ilk}a_k$, the antisymmetric part of M contributes $2|a|^2$ to $\|M\|^2$ and $4|a|^2$ to $\sum_i(\partial^j\bar{\omega}_{ji})^2$, and the two cancel identically. The Kretschmann coefficient therefore depends on all eight modes of ∂v, the Weyl-squared coefficient only on the five shear modes: the converse fails for C^2 precisely when ∂v is antisymmetric, as the example above demonstrates. Equation (42) gives the signatures $(0, 8, 0)$ and $(0, 5, 3)$.

Two consequences follow. The controlling quantity is $\partial\bar{\omega}$, not $\bar{\omega}$: a spatially uniform vorticity has $\bar{\omega} \neq 0$ with $\partial\bar{\omega} = 0$, so the order-v_s^2 curvature coefficient vanishes. A pure rigid rotation is flat; adding it to a gradient need not remove higher-order curvature. Uniform vorticity also gives $j = 0$ in Lemma 1. The leading Kretschmann exponent is $q = 2$ wherever $\partial\bar{\omega} \neq 0$. A finite-speed Kretschmann scalar is not a pure power: for the shift $F = (0, x^2, 0)$, exact computation gives $K = 4v_s^2\big(11v_s^2x^4 - 2\big)$, whose $O(v_s^2)$ coefficient is $-8 \neq 0$ as (41) requires, but which also has an $O(v_s^4)$ term and therefore vanishes at the isolated speed $v_s = \sqrt{2/11}$ on the surface $x = 1$. What $[K]^{(2)} \neq 0$ excludes is a *higher* leading exponent; the exponents fitted over a finite v_s range in Table 13(b) can include subleading corrections and motion of the maximizing point. The worst single-point relative deviation from the fitted law, over the subluminal branch the fit covers, is 1.7×10^{-2} (Alcubierre, Weyl), 1×10^{-3} (Natário, all three) and 4×10^{-15} (Rodal, all three), rising to 0.25 for Alcubierre Ricci-squared at $v_s = 0.9$, where the subleading $|S|^2$ terms of $T_{ab}T^{ab}$ are already comparable to the leading $-2|j|^2$. Those terms affect the finite-speed fit $q = 2.11$; the leading order remains v_s^2 where its coefficient is nonzero. The table reports fit quality separately

Table 9. Kinematically standardized construction diagnostics at $t = 0$, on $(N_r, N_\mu) = (32, 32), (48, 48), (64, 64)$. *(a)* Common $v_s = 0.02$, $R_c = 15\,\mathrm{m}$, $W = 4.419943\,\mathrm{m}$, and band $0.1 \le f \le 0.9$. *(b)* Reference parameter sets from Appendix D. For Garattini–Zatrimaylov, (a) uses $H = 0.02/15\,\mathrm{m}^{-1}$, $r_0 = 15\,\mathrm{m}$, $\sigma = 0.497115373\,\mathrm{m}^{-1}$; (b) uses $H = 0.1\,\mathrm{m}^{-1}$, $r_0 = 1\,\mathrm{m}$, $\sigma = 8\,\mathrm{m}^{-1}$. Both have $v = Hr_0$ and exact Type I fractions. These are parameter choices made here. Wall cells count the coarsest and finest levels. Margins are sampled Type-I slacks, with no global enclosure.

	Metric	Wall cells coarse	Wall cells fine	Type I (%)	Type IV (%)	$R_c^2 \min(\rho + p_i)$ (Type I)	WEC miss (%)	NEC miss (%)
(a)	Alcubierre	7.3	14.9	0.0	100.0	–	0.0	0.0
	Rodal	7.3	14.9	100.0	0.0	-2.7e-04	71.9	8.9
	Fuchs	7.3	14.9	100.0	0.0	0.021	–	–
	Garattini	7.3	14.9	100.0	0.0	-3.5e-04	100.0	0.0

	Metric	v_s	Wall cells coarse	Wall cells fine	Type I (%)	Type IV (%)	$R_c^2 \min(\rho + p_i)$ (Type I)	WEC miss (%)	NEC miss (%)
(b)	Alcubierre	0.5	4.5	9.2	0.6	99.4	-0.137	0.0	0.0
	Rodal	0.5	4.5	9.2	100.0	0.0	-0.194	72.6	9.6
	Fuchs	0.02	7.5	15.4	100.0	0.0	0.022	–	–
	Garattini	0.1	4.5	9.2	100.0	0.0	-0.010	100.0	0.0

Table 10. Composite diagnostic on the specified $t = 0$ slice, equations (37)–(38). NEC and finite-segment sub-scores are ratios to Alcubierre capped at one. The Type-IV column is a proper-volume wall fraction from the $N = 100$ sweep grid. Each factor is floored at 10^{-4} before forming the mean.

Metric	NEC severity	Type IV fraction	I_{seg} $\lvert \min \rvert$	Composite index
Alcubierre	1.000	0.987	1.000	0.996
Natário	1.000	0.834	1.000	0.941
Van den Broeck	1.000	0.811	0.428	0.703
Rodal	1.000	0.000	0.011	0.010

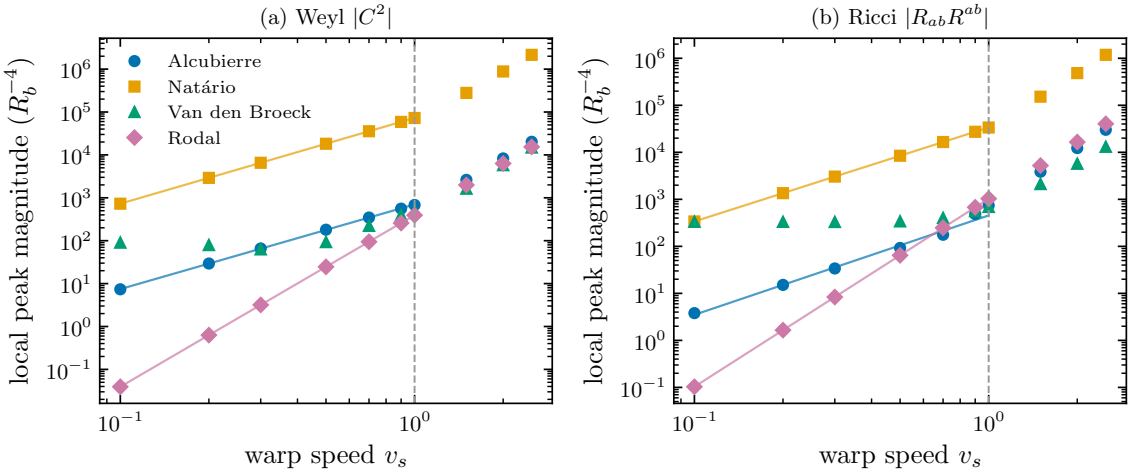

Figure 6. Local curvature maxima found from $N = 60$ wall-grid seeds ($R_b = 1$, $\sigma = 8$; log–log; $v_s = 1$ dashed). (a) Weyl-squared magnitude $|C^2|$; (b) Ricci-squared magnitude $|R_{ab}R^{ab}|$. Polishing is unconstrained and supplies no global enclosure. Lines fit the subluminal samples only where $R^2_{\log} \ge 0.9$; Van den Broeck fails this criterion. Rodal's quartic fits are consistent with the absent quadratic coefficient of its ideal irrotational profile. The finite-speed fits do not establish the coefficient identities in equations (41)–(42).

from algebraic type: curvature scalars exist for every type. The crossing of the fitted Rodal and Alcubierre Ricci curves is a comparison of those empirical curves, not a bound on either wall maximum.

4 Resolution stability

For a fixed implemented metric, automatic differentiation introduces no finite-difference step error; refining the spatial sampling grid does not change the implemented tensor at a fixed point. Roundoff and numerical profile evaluation remain relevant. Fuchs also has radial-construction discretization error, which is tested separately at fixed regularization. We assess wall extrema and volume fractions separately. All are computed on the wall-clustered graded grids of Section 3

Table 11. Speed scaling over the subluminal branch. *(a)* Free-exponent fit $|\min(\rho+p_i)| = A\,v_s^p$ of the wall NEC deficit on the Type-I branch. The Alcubierre and Natário fits use speed-dependent Type-I subsets, as discussed in the text; A is the fitted amplitude, and Van den Broeck's Type-I remainder admits no single power law ($R^2 < 0.99$). *(b)* Fixed-exponent empirical fits $\min(\rho+p_i) = -Cv_s^2$ for Alcubierre and Natário. Rodal's quadratic form follows under the $j = 0$ and fixed-domain hypotheses of equation (39); its displayed coefficient remains a numerical estimate. Here the fixed-fit statistic is $R^2 = 1 - \sum(y - Cv_s^2)^2 / \sum y^2$, with $y = |\min(\rho+p_i)|$. *(c)* Free-exponent fit of the Eulerian slice-integrated negative energy $E_- = \int_{\rho_n<0} |\rho_n|\,dV$, an Eulerian quantity throughout since $\rho_n = T_{ab}n^a n^b$ is defined at every point regardless of Hawking–Ellis type. *(d)* Locally maximized invariant magnitudes, seeded from the wall grid, fit to $X = Av_s^q$. The listed exponents and $R^2_{\log}$ describe finite-speed regressions, independently of the leading coefficient identities in the text.

(a)

Metric	exponent p	coefficient A	R^2
Alcubierre	2.03	0.634	1.0000
Natário	2.26	8.526	0.9970
Van den Broeck	no clean fit ($R^2 < 0.99$)		
Rodal	2.00	0.768	1.0000

(b)

Metric	C	dev. (all v_s)	dev. ($v_s \geq 0.5$)	R^2	NEC at sampled v_s
Alcubierre	0.631	4.09%	0.92%	1.0000	violated
Natário	7.833	86.67%	2.77%	0.9999	violated
Van den Broeck	no adequate quadratic fit ($R^2 = 0.56$)				violated
Rodal	0.768	0.00%	0.00%	1.0000	violated

(c)

	$E_- = \int_{\rho_n<0} \lvert\rho_n\rvert\,dV$	
Metric	exponent p	R^2
Alcubierre	2.00	1.0000
Natário	2.00	1.0000
Van den Broeck	0.04	0.7528
Rodal	2.00	1.0000

(d)

	Weyl $\lvert C^2\rvert$			Ricci $\lvert R_{ab}R^{ab}\rvert$		
Metric	q	A	$R^2_{\log}$	q	A	$R^2_{\log}$
Alcubierre	1.97	700.225	0.9999	2.11	454.384	0.9905
Natário	2.00	72703.741	1.0000	2.00	33751.084	1.0000
Van den Broeck	0.63	239.232	0.5362	0.18	453.277	0.5273
Rodal	4.00	393.554	1.0000	4.00	1032.108	1.0000

$(N = 80, 100, 120)$.

(i) Wall extrema, the minimum NEC and DEC margins $\min(\rho+p_i)$, $\min(\rho-|p_i|)$ and the Type-IV imaginary eigenvalue $\max|\operatorname{Im}\lambda|$, are numerically evaluated extrema. Eigenvalue degeneracies and Type-I restriction can prevent smoothness. In exact arithmetic, a sampled extremum is a one-sided bound on the corresponding infimum or supremum. Its residual gap can oscillate with grid alignment, so we assume no Richardson order and do not extrapolate it. Seeding from the deepest sample across the ladder, we instead refine the numerical tensor evaluation on a shrinking local grid, ending at a $\sim 10^{-4}$ spatial tolerance within that basin.

That refinement is *basin-local*: it cannot exclude a deeper basin missed by the grid. The polished value is a numerical estimate, at least as extreme as the computed grid values; it supplies no rigorous global bound. The separate interval search in Appendix C bounds the type-free null deficit $\mathcal{N}$ over the whole wall for all four drives, using the axisymmetric reduction. Table 15 reports each bracket and its width. The Type-I-restricted margins of the vortical walls, the dominant-condition deficits and $\max|\operatorname{Im}\lambda|$ have no global bound. *(ii) Volume fractions*, the wall Type-I/Type-IV fractions and the wall-restricted miss rates, are integrals of a discontinuous indicator. Boundary crossings can dominate their grid error; for these we report the measured stability spread across the wall-resolved ladder. The separate uniform-grid study uses the same six-direction Eulerian NEC diagnostic at $N = 25, 49, 97$; these grids put only 0.66, 1.32 and 2.64 cells across the 0.2747-wide wall. Its integrated and L^2 diagnostics are non-monotone, while the sampled minimum is unchanged to the reported precision. None of these observations establishes a convergence order or an error bound.[1]

[1] On a 41×41 window carried with the Alcubierre bubble, translation to $x_s = v_s t = 1$ moves the measured violation centroid by 1.000 and preserves the sampled violation count. The minima differ by orders of magnitude near the center, revealing numerical regularization instability. This is a check of the centroid and count only, not verification of scalar invariance by those computed minima.

Table 12. Per-diagnostic convergence and stability of the wall diagnostics on the wall-resolved graded grids ($R_b = 1$, $\sigma = 8$, $v_s = 0.5$; $N = 80, 100, 120 \to 5.9, 7.6, 8.9$ cells across the 10–90% wall). The ladder columns are grid samples and need not be monotone. Fractions are reported as ratios; their deviations are measured in percentage points. These entries are numerical diagnostics only and are not certifications.

Metric	Diagnostic	N=80	N=100	N=120	Numerical status
Alcubierre	Wall Type-IV frac.	0.989	0.987	0.990	maximum absolute deviation from the three-grid mean: 0.145 percentage points
	Wall max $\lvert\mathrm{Im}\,\lambda\rvert$	0.267	0.269	0.271	basin-local polished value: 0.272; no global enclosure
	Wall min$(\rho + p_i)$	-0.134	-0.156	-0.157	basin-local polished value: -0.158; no global enclosure
	Wall min$(\rho - \lvert p_i\rvert)$	-0.174	-0.177	-0.164	basin-local polished value: -0.196; no global enclosure
	Vorticity $\mathcal{R}_\omega$	0.333	0.333	0.333	maximum absolute deviation from the three-grid mean: 0.000 percentage points
	Wall WEC miss rate	0.000	0.000	0.000	maximum absolute deviation from the three-grid mean: 0.000 percentage points
	Wall DEC miss rate	0.000	0.000	0.000	maximum absolute deviation from the three-grid mean: 0.000 percentage points
Natário	Wall Type-IV frac.	0.859	0.834	0.845	maximum absolute deviation from the three-grid mean: 1.315 percentage points
	Wall max $\lvert\mathrm{Im}\,\lambda\rvert$	2.579	2.581	2.584	basin-local polished value: 2.588; no global enclosure
	Wall min$(\rho + p_i)$	-1.853	-1.905	-1.896	basin-local polished value: -1.955; no global enclosure
	Wall min$(\rho - \lvert p_i\rvert)$	-1.853	-1.905	-1.896	basin-local polished value: -1.955; no global enclosure
	Vorticity $\mathcal{R}_\omega$	0.508	0.505	0.506	maximum absolute deviation from the three-grid mean: 0.116 percentage points
	Wall WEC miss rate	0.000	0.000	0.000	maximum absolute deviation from the three-grid mean: 0.000 percentage points
	Wall DEC miss rate	0.000	0.000	0.000	maximum absolute deviation from the three-grid mean: 0.000 percentage points
Van den Broeck	Wall Type-IV frac.	0.869	0.811	0.839	maximum absolute deviation from the three-grid mean: 2.902 percentage points
	Wall max $\lvert\mathrm{Im}\,\lambda\rvert$	0.218	0.218	0.218	basin-local polished value: 0.219; no global enclosure
	Wall min$(\rho + p_i)$	-0.517	-0.518	-0.519	basin-local polished value: -0.521; no global enclosure
	Wall min$(\rho - \lvert p_i\rvert)$	-0.872	-0.872	-0.872	basin-local polished value: -0.873; no global enclosure
	Vorticity $\mathcal{R}_\omega$	0.440	0.439	0.440	maximum absolute deviation from the three-grid mean: 0.021 percentage points
	Wall WEC miss rate	0.588	0.548	0.574	maximum absolute deviation from the three-grid mean: 2.194 percentage points
	Wall DEC miss rate	0.561	0.513	0.531	maximum absolute deviation from the three-grid mean: 2.579 percentage points
Rodal	Wall Type-IV frac.	0.000	0.000	0.000	maximum absolute deviation from the three-grid mean: 0.000 percentage points
	Wall max $\lvert\mathrm{Im}\,\lambda\rvert$	0.000	0.000	0.000	basin-local polished value: 0.000; no global enclosure
	Wall min$(\rho + p_i)$	-0.192	-0.192	-0.190	basin-local polished value: -0.194; no global enclosure
	Wall min$(\rho - \lvert p_i\rvert)$	-0.192	-0.192	-0.190	basin-local polished value: -0.194; no global enclosure
	Vorticity $\mathcal{R}_\omega$	0.000	0.000	0.000	maximum absolute deviation from the three-grid mean: 0.000 percentage points
	Wall WEC miss rate	0.745	0.726	0.722	maximum absolute deviation from the three-grid mean: 1.408 percentage points
	Wall DEC miss rate	0.757	0.740	0.736	maximum absolute deviation from the three-grid mean: 1.271 percentage points

Per-diagnostic convergence. On the wall-resolved ladder each principal diagnostic has a measured stability spread, reported per drive (Table 12). The wall Type-IV fraction deviates from its ladder mean by at most 0.15 percentage points for Alcubierre and 1.32 for Natário, and is identically zero for Rodal, which is 100% Type I. Van den Broeck has the largest spread, 2.90 percentage points. The empirical vorticity fraction $\mathcal{R}_\omega$ is grid-stable to 0.12 percentage points or better across the ladder. The basin-local searches refine the sampled extrema without a global guarantee. The Type-IV imaginary eigenvalue peaks at $\max|\mathrm{Im}\,\lambda| = 2.588$ (Natário) down to 0 (Rodal), and the invariant NEC and DEC deficits reach the polished values tabulated. They are refined rather than extrapolated, with no assumed convergence order.

Convergence of the remaining quantities. Table 13(a) reports observed spreads of the composite across the spatial grid ladder and step refinement of the actual selected finite-segment ray. Its impact parameter and endpoints are held fixed; this checks a local numerical result and supplies no global search or tail bound. Part (b) refits the curvature exponents on the same spatial ladder. The exact scaling statements apply only under the hypotheses given above; finite-speed exponents and grid stability remain empirical. Parts (c)–(e) give the native Rodal and observer-search checks.

Table 13. Resolution stability of the diagnostics that admit no Richardson order. *(a)* Composite diagnostic on the wall-resolved grid ($N = 80/100/120$) and the selected finite-segment ray at the listed step densities per $16/s$, with fixed impact parameter and endpoints. Grid "stable" means a range at most $\max\{0.02, 0.05(\overline{\mathcal{E}} + 10^{-12})\}$; ray "stable" means a finest-pair change at most $10^{-8} + 10^{-4}|I_{\text{finest}}|$. *(b)* Fitted curvature-invariant exponents $X_{\max} = A\, v_s^q$ ($R_b = 1$, $\sigma = 8$), refit at each level; poor power-law fits are marked separately. *(c)* Rodal missed fractions at the native parameters ($v_s = 0.5$, $R_b = 100$, $\sigma = 0.03$, $N_{\text{starts}} = 8$) across uniform grids; all points are Type I at $N \geq 50$, so the reference criterion is the algebraic slack. *(d)* N_{starts} ablation at $v_s = 0.5$ on a 50^3 grid ($\zeta_{\max} = 5$). *(e)* Alcubierre at $v_s = 0.5$, $R_b = 1$, $\sigma = 8$, $t = 0$, on uniform $N = 25, 49, 97$ grids in $[-5, 5]^3$. All levels use the minimum m_6 of the six axis-aligned Eulerian null contractions. The integrated and L^2 diagnostics sum its negative part and squared negative part with coordinate cell-volume weights, taking a square root for the latter. The coordinate center is excluded. The last column is the maximum absolute departure from the three-grid mean, with no extrapolation or assumed order.

(a)

	Composite diagnostic (N)				Selected finite ray	
Metric	80	100	120	verdict	step density: integral	verdict
Alcubierre	0.996	0.996	0.997	stable (0.0008)	16384: −0.146227 32768: −0.146227 65536: −0.146227	stable ($\delta = 8.7e-14$)
Natário	0.951	0.941	0.945	stable (0.0092)	16384: −0.540367 32768: −0.540367 65536: −0.540367	stable ($\delta = 1.5e-09$)
Van den Broeck	0.719	0.703	0.711	stable (0.0162)	16384: −0.062634 32768: −0.062634 65536: −0.062634	stable ($\delta = 5.0e-13$)
Rodal	0.010	0.010	0.010	stable (0.0000)	16384: −0.001671 32768: −0.001671 65536: −0.001671	stable ($\delta = 9.0e-14$)

(b)

Metric	Invariant	fitted q N=80	N=100	N=120	Spread
Alcubierre	Weyl $\lvert C^2\rvert$	1.97	1.97	1.97	0.00
	Ricci $\lvert R_{ab}R^{ab}\rvert$	2.11	2.11	2.11	0.00
Natário	Weyl $\lvert C^2\rvert$	2.00	2.00	2.00	0.00
	Ricci $\lvert R_{ab}R^{ab}\rvert$	2.00	2.00	2.00	0.00
Van den Broeck	Weyl $\lvert C^2\rvert$	poor power-law fit ($R^2_{\log} < 0.99$)			
	Ricci $\lvert R_{ab}R^{ab}\rvert$	poor power-law fit ($R^2_{\log} < 0.99$)			
Rodal	Weyl $\lvert C^2\rvert$	4.00	4.00	4.00	0.00
	Ricci $\lvert R_{ab}R^{ab}\rvert$	4.00	4.00	4.00	0.00

(c)

	f_{miss} (%)				Total violated (%)			
N	NEC	WEC	SEC	DEC	NEC	WEC	SEC	DEC
25	1.8	15.7	28.4	28.4	87.3	87.3	87.3	100.0
50	1.6	15.6	28.0	28.5	87.0	87.0	87.0	99.9
100	1.5	15.6	27.8	28.7	86.9	86.9	86.9	99.9

(d)

	N_{starts}				
Metric	1	2	4	8	16
Min robust WEC margin†					
Alcubierre	−3460.2	−3460.2	−3460.2	−3460.2	−3460.2
Rodal	-2.6×10^{-6}	-2.6×10^{-6}	-2.6×10^{-6}	-2.6×10^{-6}	-2.6×10^{-6}
Missed WEC (%)					
Alcubierre	0.00	0.00	0.00	0.00	0.00
Rodal	15.60	15.60	15.60	15.60	15.60

†Robust margin: algebraic slack at Type I points; capped extremum at non-Type I points.

(e)

Quantity	$N = 25$	$N = 49$	$N = 97$	$\max_i \lvert Q_i - \overline{Q}\rvert$
Sampled min NEC	−0.631	−0.631	−0.631	0
Integrated viol.	1.590	1.721	1.652	0.066
L^2 viol. norm	0.75	0.77	0.73	0.020

4.1 Curvature identity validation

As an independent check of the autodiff curvature chain, the computed Riemann tensor satisfies the algebraic symmetries and the first Bianchi identity to residual $< 10^{-10}$, and the contracted Bianchi identity $\nabla_a G^{ab} = 0$ to $< 10^{-6}$ (third-derivative cancellation) at representative Schwarzschild and Alcubierre points; $T_{ab} = G_{ab}/8\pi$ is symmetric to a relative $< 10^{-13}$.

4.2 Optimization stability

To measure sensitivity to the multi-start budget, we vary the number of restarts $N_{\text{starts}} \in \{1, 2, 4, 8, 16\}$ at $v_s = 0.5$ for Alcubierre and Rodal, recomputing only the observer search on a fixed curvature grid with independent (non-nested) random draws per setting. As Table 13(d) shows, the minimum margins and missed-violation fractions are insensitive to the multi-start budget: Alcubierre and Rodal are N_{starts}-independent from a single start (Rodal, 100% Type I, bypasses the optimizer entirely). The default observer-search budget is $N_{\text{starts}} = 8$; the matched benchmark of Table 7(b) uses $N_{\text{starts}} = 2$. This ablation covers Alcubierre and Rodal only and does not establish optimization stability for the other two drives.

5 Discussion

Single-frame (Eulerian) evaluation can systematically understate the energy-condition structure of warp drive spacetimes, and the way it does so is geometry-dependent. For the everywhere-Type I Rodal metric, presented in [6] as predominantly positive in invariant energy density while still violating the null energy condition, the Eulerian diagnostic misses most wall weak- and dominant-energy violations (Section 3.1); the Type-I inequalities define the exact pointwise test, while the reported fractions and ablation spreads remain numerical (Appendix B). The sampled Alcubierre and Natário walls also contain Type-IV regions, where a rest-frame energy density does not exist. Van den Broeck is intermediate (WEC 55%, DEC 51% wall miss at $v_s = 0.5$). The contraction magnitude also depends on the observer. At NEC-violating Type I points it tends to $-\infty$ along suitable timelike boosts, with $\gamma^2 \simeq e^{2\zeta}/4$ (Section 2.5). The algebraic energy-condition verdict requires no rapidity cap.

Physical interpretation. In a Type-I rest frame the WEC density along a principal spatial direction is $\rho + (\rho + p_i)\sinh^2\zeta$. If $\rho > 0$ and $\rho + p_i < 0$, it becomes negative when $\sinh^2\zeta > \rho/|\rho + p_i|$. The angle of that spatial direction is measured within the rest frame; an Eulerian rapidity cap is a different restriction unless the two frames coincide. The DEC uses the separate energy-flux test in Section 2.5, and its Type-I verdict follows directly from $\rho \geq |p_i|$.

Scope of the physical conclusion. Energy-condition satisfaction requires a criterion covering every admissible causal vector. The Type-I eigenvalue inequalities and the LMI reductions provide such criteria. In particular, positive Eulerian energy density does not ensure satisfaction, while an established Type-IV tensor violates the NEC and hence the other three conditions.

The no-go results of Santiago–Schuster–Visser [2] apply under their stated geometric and subsidiary hypotheses. Our calculation concerns the stress-energy induced by each specified metric. Matter realizability, global causal properties, and curved-space quantum inequalities require additional analysis. The finite-segment and composite diagnostics retain the normalization and sampling limits stated in Sections 1.2 and 3.6.

6 Conclusion

The classical S-lemma gives pointwise tests of all four energy conditions without a Hawking–Ellis type assumption or rapidity cap. Interval evaluation of the metric and curvature connects these algebraic criteria to the induced stress-energy. At the reference parameters, the global bounds establish a negative normalized null-energy infimum in each of the four benchmark walls. With the specified Eulerian normalization, the bounds place Natário and Van den Broeck below Alcubierre; they do not order the other pairs.

The momentum constraint explains why an irrotational shift on time-independent Euclidean slices with unit lapse gives Type-I stress-energy. For smooth shifts on all of $\mathbb{R}^3$ with bounded vorticity, momentum vanishes identically precisely for a gradient plus rigid rotation. The momentum-plane type criterion also requires the stated stress-axis splitting hypothesis. The ideal Rodal profile is Type I everywhere, yet the Eulerian diagnostic misses about 73% of its sampled wall WEC violations and 74% of its DEC violations. These fractions and the other drives' vorticity/type association remain numerical results for the specified slices and masks.

For speed-linear shifts on fixed Euclidean slices, integrated negative Eulerian energy scales exactly as v_s^2 when finite. A quadratic pointwise eigenvalue-margin law follows when $j = 0$, with a fixed comoving profile and domain. The leading Kretschmann coefficient depends on all spatial derivatives of shift vorticity, while the Weyl-squared coefficient has a nontrivial kernel. Finite-speed regressions remain distinct from these coefficient identities.

The regularized Fuchs and Garattini–Zatrimaylov panel compares kinematic choices only. The null-geodesic integrals are finite-segment, basin-local diagnostics with selected-ray step refinement; they supply no complete-geodesic ANEC conclusion. The flat-space quantum-inequality estimates likewise supply no curved-spacetime bound.

The matrix criteria apply to any symmetric stress-energy tensor on a four-dimensional Lorentzian spacetime. WARPAX implements them for prescribed metrics, making the distinction between positive energy for one observer and satisfaction of an energy condition explicit. Matter realizability and global causal viability require additional analysis.

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Data availability

WARPAX is freely available under the MIT license at https://github.com/anindex/warpax. The source and numerical data support the calculations reported here. Archived releases are available at https://doi.org/10.5281/zenodo.18715933. No external datasets are required.

A Numerical implementation

WARPAX is a Python package built on JAX [11] and Equinox [12]. Each metric is a map $x^\mu \mapsto g_{ab}(x)$: the four analyzed here (Alcubierre [1], Natário [23], Van den Broeck [24], Rodal [6]) and two benchmarks (Minkowski, and Schwarzschild in isotropic Cartesian coordinates), with the full ADM decompositions in Appendix D. Derivatives along the curvature chain are taken by forward-mode automatic differentiation, so the derivatives of the implemented metric are exact up to roundoff. The JAX calculations use 64-bit floating point; interval calculations use the precision stated in Appendix E.2.

Metric regularity. The warp metrics use C^∞ tanh-based shaping. Radial expressions divide by $\sqrt{r_s^2 + \varepsilon^2}$ to remain C^∞ at the center, with $\varepsilon^2 = 10^{-60}$ in the shape and angular profiles and the coarser $\varepsilon^2 = 10^{-12}$ in the divisor that builds the unit radial direction; the residual curl from the coarser regularization is quantified in Appendix B. Curvature is computed from derivatives of the implemented profile.

Pointwise evaluation. For a given metric and grid we evaluate the curvature chain $g_{ab} \to T_{ab}$, classify each point by Hawking–Ellis type (I, II, III or IV), compute the algebraic slacks (14)–(17) at Type I points, use the multi-start observer optimization for non-Type-I diagnostics (Section 2.5), and report the verdicts. At Type I the algebraic slacks decide and at Type IV the absence of a causal eigenvector forces the null violation; the capped extrema report severity only. Every non-Type-I point, whatever its label, takes its Boolean verdict from Theorem 1. The Eulerian baseline (six axis-aligned null rays for the null condition, the ADM normal for the timelike ones) is, for any fixed observer u_0, an upper bound on the worst-case margin, $m^*(x) \le m(x, u_0)$, so its violation set lower-bounds the true extent; that inequality is enforced as a consistency check.

B Rodal DEC ablation study

Regularization of the coordinate origin. The Rodal metric is implemented with two C^∞ regularizations $\sqrt{r^2 + \varepsilon^2}$ of the coordinate radius at $r = 0$: the shape and angular profiles take $\varepsilon^2 = 10^{-60}$, and the divisor that builds the unit radial direction takes the coarser $\varepsilon^2 = 10^{-12}$, without which $\partial_i n_j$ diverges at the bubble center. The coarser regularization breaks exact irrotationality: replacing x^i/r by $x^i/\sqrt{r^2 + \varepsilon^2}$ leaves a residual curl linear in ε^2. At the sampled wall points, $|\nabla \times \beta|/|\beta|$ is at most 10^{-10} in the coordinates used, ten orders below the $O(1)$ values of the vortical drives. Lemma 3 proves Type I for the ideal analytic profiles. It does not establish the type of the separately regularized implementation; the sampled type labels and residual-curl

Table 14. Rodal DEC ablation. *(a)* Resolution and shape-function regularization stability of the Rodal grid DEC miss rate at fixed wall thickness ($\sigma = 0.03$); the rate is insensitive to both, and the Alcubierre control misses 0.0% throughout. *(b)* Wall-thickness sweep, wall-resolved across its ladder. Every σ is evaluated on the exact axisymmetric (r, μ) reduction, on a four-level ladder $(N_r, N_\mu) = (48, 48)$, $(64, 64)$, $(80, 80)$, $(96, 96)$ with proper-volume weights $2\pi r^2$. Both DEC miss columns are conditional on violating volume: "Grid" uses the full sampled ball and "Wall" restricts to $0.1 \le f \le 0.9$. "Wall cells" is the worst-case count across the 10–90% wall normal at the coarsest and finest level. The quoted spread is the largest departure from the finest level across the whole ladder, not the gap between the two finest, which at $\sigma = 0.1$ would understate the drift by a factor of fifty.

(a)

Sweep	Parameter value	Rodal grid DEC miss (%)	Alcubierre grid DEC miss (%)
Resolution (N)	25	28.36	0.0
	50	28.53	0.0
	100	28.68	0.0
Regularization (ε^2)	10^{-24}	28.53	–
	10^{-18}	28.53	–
	10^{-12}	28.53	–
	10^{-6}	28.53	–

(b)

	Wall cells		Wall	Grid DEC	Wall DEC
σ	coarsest	finest	points	miss (%)	miss (%)
0.01	26.1	52.4	5,856	46.98	52.88 ± 0.42
0.03	13.0	25.7	2,976	34.46	63.16 ± 0.78
0.1	7.5	15.5	2,016	32.10	76.78 ± 1.92
0.3	6.3	11.7	1,920	30.67	85.46 ± 0.53

checks supply numerical evidence for that implementation. A controlled ablation with three single-variable sweeps (resolution, shape-function regularization, wall thickness; Alcubierre as control) shows the $\sim$28.5% DEC miss rate is insensitive to resolution (0.32 pp) and to ε^2 in the shape function (0.0 pp across 10^{-24} to 10^{-6}). The separate wall-thickness study compares proper-volume-weighted conditional rates over the full domain and within the wall, as defined below; the Alcubierre control misses 0.0% throughout.

The sweeps are run at baseline $v_s = 0.5$, $N = 50$, $\sigma_{\text{Rodal}} = 0.03$ and $\varepsilon^2 = 10^{-24}$ in the swept shape function, with the direction divisor held at its production value 10^{-12}. Table 14(a) reports the resolution and regularization sweeps. The third, over wall thickness, runs $52.9 \to 62.7 \to 75.4 \to 87.3\%$ in the wall-restricted conditional rate on the uniform grid, whose two sharpest σ span 1.79 and 0.598 cells and so fall below the four-cell floor; the wall-resolved values quoted in the text are $52.9 \to 63.2 \to 76.8 \to 85.5\%$ (Table 14(b)). The trend agrees in direction and shape on both, but quantitative conclusions use the wall-resolved sweep.

In the wall-thickness study, the full-domain conditional DEC miss rate falls from 47.0% to 30.7% as the wall sharpens. This is the proper-volume-weighted fraction of DEC violations missed in the full sampled ball, distinct from the unconditional Cartesian-grid fraction in equation (34). Restricting both numerator and denominator to violating wall points gives the wall rate, which rises from 52.9% to 85.5% (Table 14(b)). Thus the two sampled regions show different trends, and the Eulerian miss persists within the wall itself. No miss is detected for the Alcubierre control.

The trend is measured on a resolved wall. The uniform $N = 50$ grid falls below the four-cell criterion at the two sharpest walls, and reaching four cells at $\sigma = 0.3$ on a uniform Cartesian grid would require $N \ge 329$, that is $\sim 3.6 \times 10^7$ tensor evaluations. The exact axisymmetric reduction costs a few thousand points for the same physics, so every σ spans at least 6.3 cells at the coarsest ladder level and the conditional rate is converged to better than 0.3 percentage points between the two finest levels. Against the uniform grid the endpoints move by at most 2 points and the rise is monotone throughout.

C Global enclosures of the wall null deficit

A local search can find a negative null contraction but cannot exclude a deeper minimum elsewhere. To bound the global wall infimum, we cover the whole wall with interval bounds and refine the regions that could contain the minimum.

Exact dimensional reduction. The four matched Alcubierre, Natário, Van den Broeck and Rodal metrics are invariant under rotations about the propagation axis. Their Eulerian normal and radial wall mask share this symmetry, so the Eulerian-normalized null minimum defined below depends only on (x, s) with $s = (y^2 + z^2)^{1/2}$. Each orbit has a representative with $z = 0$, $y = s \ge 0$; restricting this objective and mask to that half-plane is therefore exact.

The bounded objective. We bound the null-energy deficit in the form that is well defined at every point irrespective of algebraic type. Writing a future null direction as $k^a = n^a + v^a$ with v^a orthogonal to n^a and of unit length in the induced metric,

$$q(v) = T_{ab}k^a k^b = \rho_n - 2\,b\cdot v + v^{\mathsf{T}} S v, \qquad b_i = -T_{ab}n^a e_i^b, \quad S_{ij} = T_{ab}e_i^a e_j^b, \tag{43}$$

and the pointwise deficit is $\mathcal{N}(x) = \min_{|v|=1} q(v)$, whose sign decides the null energy condition at x. At a point with vanishing Eulerian momentum, $b = 0$ and $\mathcal{N} = \min_i(\rho + p_i)$. This identity applies to the ideal irrotational Rodal profile. The enclosed implementation uses the radial regularizations of Appendix B, so its bound concerns $\mathcal{N}$ directly and does not assume exact equality with an eigenvalue slack.

Method. The bounds combine interval differentiation, null-vector evaluation, and a matrix lower bound. Interval forward-mode differentiation through second metric derivatives encloses the curvature chain $g \to \Gamma \to R^a{}_{bcd} \to R_{ab} \to G_{ab} \to T_{ab}$. Repeated occurrences of the same variable can widen these enclosures; centered mean-value bounds reduce this dependency overestimation as described below. The only transcendental in the shape function is tanh, and $\tanh u = 1 - 2/(e^{2u}+1)$ contains u exactly once, so its interval evaluation is free of dependency overestimation. Perturbing the Eulerian data changes $\mathcal{N}$ by at most $|\delta\rho_n| + 2|\delta b| + \|\delta S\|_2$. Thus $\mathcal{N}$ is 1-Lipschitz with respect to this weighted norm.

To obtain an upper bound, a floating-point secular solve proposes a spatial direction at a point whose wall membership is established. We normalize that direction and evaluate (43) in interval arithmetic. Any such null contraction bounds the wall infimum from above; a suboptimal direction weakens the bound but preserves its validity. The lower bound combines two independent inequalities valid throughout each box, as described under *Coverage* below.

The search uses a Moore–Skelboe branch-and-bound: a priority queue of boxes ordered by the lower end of the enclosure, splitting the most promising box and discarding boxes that cannot contain the global minimum. On exit

$$\mathcal{N}_{\text{lower}} \;\le\; \inf_{\text{wall}} \mathcal{N} \;\le\; \mathcal{N}_{\text{upper}}, \tag{44}$$

with the lower end valid on the *continuum* rather than at sample points. A box is dropped from the wall mask only when its shape-function enclosure lies entirely outside $f \in [0.1, 0.9]$, so no wall point is dropped, including points on the curved mask boundary; and a box straddling the coordinate axis $r_s = 0$, where the spherical form carries a removable $1/r_s$, is reported as unbounded and split. The exterior of the search box is excluded by a single interval evaluation. The wall restriction $0.1 \le f \le 0.9$ is a condition on the shape function alone, and f depends on position only through $r_s = |\mathbf{x} - \mathbf{x}_s(t)|$. The top hat is strictly decreasing in r_s for $r_s > 0$, since $\mathrm{d}f/\mathrm{d}r_s \propto \operatorname{sech}^2[\sigma(r_s+R)] - \operatorname{sech}^2[\sigma(r_s-R)] < 0$ there, so one interval evaluation at $r_s = 3$ bounds f on the whole unbounded exterior by $[0,\, 1.3 \times 10^{-14}]$, disjoint from the band. The exterior therefore contains no wall point in *any* direction, $x < -3$ included.

Result. Table 15 reports the two ends of (44).

At the reported search point x_* we separately enclose the pointwise deficit $\mathcal{N}(x_*)$. Its midpoint is printed as an achieved numerical value, with the local interval width. This error at a fixed point is distinct from uncertainty in the global infimum: the latter lies in the full bracket $[\mathcal{N}_{\text{lower}}, \mathcal{N}_{\text{upper}}]$. The upper endpoint is a rigorous bound obtained from an evaluated null contraction, and need not equal an exactly attained minimum. Agreement with basin-local polishing does not reduce the uncertainty in the global infimum.

The lower endpoint is valid over the whole wall. Interval dependency and the finite search budget can leave a wide global bracket even when the local point evaluation is sharp. The table reports both scales.

Coverage. The interval calculation covers all four drives, including Van den Broeck's conformally flat spatial metric $\gamma_{ij} = B^2\delta_{ij}$. All four searches yield finite brackets (Table 15).

The full enclosures establish a negative wall infimum for every drive. Natário and Van den Broeck each have a strictly lower infimum than Alcubierre. The other pairs have overlapping brackets and are not ordered by these bounds. These comparisons use the specified Eulerian normalization and slice. Every search exhausts its evaluation budget before reaching the chosen target interval width 10^{-4}.

Table 15. Rigorous global search for the wall null-energy deficit $\mathcal{N} = \min_{|v|=1} q(v)$ at matched parameters ($R_b = 1$, $\sigma = 8$, $v_s = 0.5$), by interval branch-and-bound on the axisymmetry-reduced domain. Evaluation budgets are 120,000 for Alcubierre and 4096 for the other drives; the target width 10^{-4} is not reached. $\mathcal{N}(x_*)$ is the midpoint of a separate point enclosure, whose width gives the local numerical uncertainty; $\mathcal{N}_{\text{upper}}$ bounds the global infimum from above; $\mathcal{N}_{\text{lower}}$ is a continuum bound over the wall band $f \in [0.1, 0.9]$. $\mathcal{N}$ includes all algebraic types; the sampled Type-I margins are reported separately.

Metric	$\mathcal{N}(x_*)$	$\mathcal{N}_{\text{lower}}$	$\mathcal{N}_{\text{upper}}$	local interval width
Alcubierre	-0.631077	-0.631624	-0.631076	$2.22e-16$
Rodal	-0.194141	-23.496805	-0.194141	$1.07e-14$
Natário	-5.212534	-1419.173790	-5.212534	$1.78e-15$
Van den Broeck	-1.281561	-35.315137	-1.281561	$8.88e-16$

The lower bound is the larger of two independently valid bounds. The first is $\rho_{\text{lo}} - 2|b|_{\max} + \lambda_{\min,\text{lo}}(S)$, with the spatial eigenvalue bounded by the Gershgorin inequalities in interval arithmetic. It bounds the linear and quadratic terms separately and need not become sharp as a box shrinks. The second uses $\mathcal{N} \geq 2\lambda_{\min}(M(\sigma))$ for any σ, with the eigenvalue bounded by an interval LDL^{T} test. Refinement can tighten this bound when the interval pivots remain valid; multiplier choice and interval overestimation both leave slack. Corollary 2 gives equality for an exact best multiplier and exact eigenvalue.

Where interval gradients are available, each orthonormal stress component $h \in \{\rho_n, b_i, S_{ij}\}$ also has the centered mean-value enclosure

$$h(B) \subseteq h(c) + \partial_x h(B)(B_x - c_x) + \partial_s h(B)(B_s - c_s),$$

where B is a box in (x, s) and c its center. Differentiating the stress components requires third metric derivatives on the box; the four enclosed wall profiles are smooth there. Both lower bounds are evaluated with these enclosures as well, and the largest valid bound is retained. If the tightening cannot be evaluated, the ordinary interval bounds remain valid. The objective uses orthonormal spatial components and contractions with the Eulerian normal. Spatial coordinate components coincide with that orthonormal basis when $\gamma_{ij} = \delta_{ij}$; time and flux components still require the normal. On Van den Broeck's conformally flat slice the chain Cholesky-factors $\gamma = LL^{\mathsf{T}}$ to construct the orthonormal frame.

The bracket bounds $\mathcal{N}$ on the continuum of the wall region at fixed physical parameters. Note that it is a statement about $\mathcal{N}$ and not about the Hawking–Ellis type assignment, which remains numerically classified and tolerance-dependent (Section 1.2).

D Metric definitions

For reproducibility, we collect the ADM $3+1$ decompositions of the test metrics. In each case the line element takes the general form

$$\mathrm{d}s^2 = -\alpha^2\,\mathrm{d}t^2 + \gamma_{ij}\big(\mathrm{d}x^i + \beta^i\,\mathrm{d}t\big)\big(\mathrm{d}x^j + \beta^j\,\mathrm{d}t\big), \tag{45}$$

where α is the lapse, β^i the shift vector, and γ_{ij} the spatial metric. The four benchmark metrics share the bubble center $x_s(t) = v_s\,t$ and the comoving distance $r_s = \sqrt{(x - x_s)^2 + y^2 + z^2}$; the Garattini–Zatrimaylov bubble instead moves radially with $r(t) = r_0 e^{Ht}$ and $v = Hr(t)$, as set out below. The Fuchs shell uses stationary comoving coordinates and $r = |\mathbf{x}|$. The standard tanh-smoothed top-hat shape function is

$$f(r_s) = \frac{\tanh\big[\sigma(r_s + R_b)\big] - \tanh\big[\sigma(r_s - R_b)\big]}{2\,\tanh(\sigma R_b)}, \tag{46}$$

with $f(0) = 1$ and $f(r_s) \to 0$ for $r_s \gg R_b$.

Alcubierre. $\alpha = 1$, $\gamma_{ij} = \delta_{ij}$, $\beta^i = (-v_s\,f(r_s),\ 0,\ 0)$. The shape function f is given by equation (46) [1].

Van den Broeck. $\alpha = 1$, $\beta^i = (-v_s\,f(r_s),\ 0,\ 0)$, $\gamma_{ij} = B^2(r_s)\,\delta_{ij}$, where $B(r_s) = 1 + \alpha_{\text{vdb}}\,f_B(r_s)$. The inner shape function f_B uses a separate radius $\tilde{R}$ and width σ_B [24]:

$$f_B(r_s) = \frac{\tanh\big[\sigma_B(r_s + \tilde{R})\big] - \tanh\big[\sigma_B(r_s - \tilde{R})\big]}{2\,\tanh(\sigma_B \tilde{R})}. \tag{47}$$

Natário. $\alpha = 1$, $\gamma_{ij} = \delta_{ij}$. Natário writes the rest-frame metric as $\mathrm{d}s^2 = -\mathrm{d}t^2 + |\mathrm{d}\boldsymbol{\xi} - X(\boldsymbol{\xi})\mathrm{d}t|^2$, with $X(0) = 0$ and $X(\infty) = -v_s\mathbf{e}_x$ [23]. Under $\mathbf{x} = \boldsymbol{\xi} + v_s t\mathbf{e}_x$, the $+\beta$ laboratory shift is $\beta_{\mathrm{lab}} = -X(\mathbf{x} - v_s t\mathbf{e}_x) - v_s\mathbf{e}_x$, hence

$$\begin{aligned}\beta^x &= v_s[2n(r_s) + r_s n'(r_s)\sin^2\theta - 1],\\ \beta^y &= -v_s n'(r_s)\frac{(x - v_s t)y}{r_s}, \qquad \beta^z = -v_s n'(r_s)\frac{(x - v_s t)z}{r_s}.\end{aligned} \tag{48}$$

Here $n = (1 - f)/2$, with f from equation (46), and $\sin^2\theta = (y^2 + z^2)/r_s^2$. This analytic profile approaches the constant interior/exterior profiles used in Ref. [23]; it is not compactly supported. The origin uses the analytic even extension in r_s^2. The shift is divergence-free, tends to zero at infinity, and equals $-v_s\mathbf{e}_x$ at the center. Thus the center worldline $C(t) = (t, v_s t, 0, 0)$ has $g(\dot{C}, \dot{C}) = -1$ for every speed. The rest/laboratory change leaves the slices $t =$ constant unchanged; the normal is the same geometric observer expressed in different coordinates. All numerical results here use the laboratory form.

Rodal. $\alpha = 1$, $\gamma_{ij} = \delta_{ij}$. The irrotational (curl-free) shift is specified in a spherical tetrad aligned with the propagation axis (θ measured from $+x$) [6]:

$$\hat{\beta}_r = -v_s\, F(r_s)\cos\theta\,, \qquad \hat{\beta}_\theta = v_s\, G(r_s)\sin\theta\,, \tag{49}$$

where $F = f$ (equation 46) and

$$G(r_s) = 1 - \frac{2\,r_s\sigma\sinh(R_b\sigma) + \cosh(R_b\sigma)\big[\ln\cosh\sigma(r_s - R_b) - \ln\cosh\sigma(r_s + R_b)\big]}{2\,r_s\sigma\sinh(R_b\sigma)}\,. \tag{50}$$

Note on conventions (Rodal). Ref. [6] presents the irrotational construction in a Natário-style "bubble-at-rest" convention, with radial profiles $f_{\mathrm{paper}}(0) = g_{\mathrm{paper}}(0) = 0$ and $f_{\mathrm{paper}}(\infty) = g_{\mathrm{paper}}(\infty) = 1$. We standardize to a lab-frame convention with vanishing shift at infinity:

$$F(r_s) = 1 - f_{\mathrm{paper}}(r_s), \qquad G(r_s) = 1 - g_{\mathrm{paper}}(r_s), \tag{51}$$

so that $F(0) = G(0) = 1$ and $F(\infty) = G(\infty) = 0$. Subtracting a uniform asymptotic translation preserves irrotationality (a uniform translation field is curl-free). We use the "$+\beta$" ADM convention $ds^2 = -\alpha^2\,dt^2 + \gamma_{ij}(dx^i + \beta^i\,dt)(dx^j + \beta^j\,dt)$; Ref. [6] uses "$-\beta$", so component-level comparison requires the corresponding sign flip.

Removable $r_s \to 0$ form in $G(r_s)$. The apparent $1/r_s$ factor in equation (50) is a removable $0/0$ form of the spherical-basis representation. Defining $\Delta(r_s) \equiv \ln\cosh\big[\sigma(r_s - R_b)\big] - \ln\cosh\big[\sigma(r_s + R_b)\big]$, one finds $\Delta'(0) = -2\sigma\tanh(\sigma R_b)$, whence $\lim_{r_s\to 0} G(r_s) = 1$. The irrotational construction is therefore regular at the bubble center; the $r_s \mapsto \sqrt{r_s^2 + \varepsilon^2}$ substitution is a numerical stability device for automatic differentiation, not a physical repair.

Cartesian shift components. In Cartesian coordinates, equations (49) become

$$\boldsymbol{\beta} = -v_s\big[G(r_s)\,\hat{\boldsymbol{x}} + \big(F(r_s) - G(r_s)\big)\,n_x\,\boldsymbol{n}\big], \tag{52}$$

where $\boldsymbol{n} = (\Delta x,\, y,\, z)/r_s$ is the radial unit vector from the bubble center and $n_x = \Delta x/r_s$. The supplied profiles are even analytic functions near $r_s = 0$, with $F(0) = G(0) = 1$ and $F - G = O(r_s^2)$. Thus $(F - G)/r_s^2$ is smooth in r_s^2, and $(F - G)n_x n_i = [(F - G)/r_s^2]\Delta x\,(x_i - x_{s,i})$ extends smoothly to the center. The shift there is $\boldsymbol{\beta}(0) = -v_s\,\hat{\boldsymbol{x}}$.

Fuchs constant-velocity shell. Fuchs *et al.* [7] begin with a static matter-shell construction before modifying the metric. A shell of constant density occupies $R_1 \le r \le R_2$ with total mass M; the Tolman–Oppenheimer–Volkoff equation

$$\frac{\mathrm{d}p}{\mathrm{d}r} = -\frac{(\rho + p)\,(m + 4\pi r^3 p)}{r\,(r - 2m)}, \qquad p(R_2) = 0, \tag{53}$$

is integrated inward for the isotropic pressure; density and pressure are then smoothed by four passes of a moving average with width ratio $s_\rho/s_P \simeq 1.72$, the cumulative mass $m(r)$ recomputed from the smoothed density, and the metric potentials obtained from

$$e^{2b(r)} = \left(1 - \frac{2m}{r}\right)^{-1}, \qquad \frac{\mathrm{d}a}{\mathrm{d}r} = \frac{m + 4\pi r^3 p}{r\,(r - 2m)}, \tag{54}$$

with the Schwarzschild boundary condition $e^{2a} = e^{-2b} = 1 - 2M/r$ outside the solved grid, which we impose analytically rather than by extrapolating the numerical solution. Following equation (26) of Ref. [7], the warp modification changes only covariant $g_{01} = g_{10}$:

$$g_{00} = -e^{2a(r)}, \qquad g_{0x} = -v_s S_{\mathrm{warp}}(r), \qquad g_{0y} = g_{0z} = 0, \qquad g_{ij} = \gamma_{ij} = \delta_{ij} + (e^{2b(r)} - 1) n_i n_j. \tag{55}$$

Here $n_i = x_i/r$ in stationary comoving coordinates. The compatible ADM quantities are $\beta^i = -v_s S_{\mathrm{warp}} \gamma^{ix}$ and $\alpha^2 = e^{2a} + v_s^2 S_{\mathrm{warp}}^2 \gamma^{xx}$. For positive-definite γ this lapse is positive and $\det g = -\alpha^2 \det \gamma < 0$.

We replace the published buffered clamps by the natural-endpoint extension

$$S_{\mathrm{warp}}(r) = \begin{cases} 1, & r \le R_1, \\ [1 + \exp(-\chi(r))]^{-1}, & R_1 < r < R_2, \\ 0, & r \ge R_2, \end{cases} \qquad \chi = (R_2 - R_1)\left(\frac{1}{r - R_2} + \frac{1}{r - R_1}\right). \tag{56}$$

All derivatives of $S_{\mathrm{warp}} - 1$ vanish at R_1, and all derivatives of S_{warp} vanish at R_2. This C^∞ extension is the manuscript's regularization of the published prescription. The extension uses the physical shell endpoints.

The numerical radial potentials use a cubic interpolant with not-a-knot boundary conditions; its exact spline equations enforce C^2 continuity. To join it to constant interior potentials $(a_0, 0)$ and analytic Schwarzschild exterior potentials, we blend with $Q(u) = 6u^5 - 15u^4 + 10u^3$, whose first two derivatives vanish at both endpoints. The inner blend occupies $[R_1/4, R_1/2]$; the outer blend occupies $[(R_2 + r_{\max})/2, r_{\max}]$, with $r_{\max} = 1.5R_2$. These joins are also this manuscript's regularization. In exact arithmetic the metric is at least C^2, including its center and all interpolation and matching radii. Stored binary64 spline coefficients leave second-derivative matching residuals of at most 1.46×10^{-14} on the tested radial grids.

The published parameters are $R_1 = 10\,\mathrm{m}$, $R_2 = 20\,\mathrm{m}$, $M = 4.49 \times 10^{27}\,\mathrm{kg}$ ($2M = 6.668692\,\mathrm{m}$), and $v_s = 0.02$. We use four smoothing passes with fixed physical density smoothing width $\sigma_\rho = 0.06(R_2 - R_1)$ and pressure width $\sigma_P = \sigma_\rho/1.72$. The cell-integrated moving-average kernel has span $\sqrt{12}\sigma$; these widths and all joins are held fixed when the radial grid is refined. Production uses 2048 radial points. Curvature probes use 512, 1024, 2048 and 4096 points; the construction-panel diagnostics use 1024, 2048 and 4096. The sigmoid's 10–90% crossings remain 12.790029 and 17.209971 m, giving $R_c = 15\,\mathrm{m}$ and $W = 4.419943\,\mathrm{m}$.

Garattini–Zatrimaylov de Sitter bubble. The construction of [10] is a shift-only metric on *flat* slices (de Sitter in Painlevé–Gullstrand form, with the expansion carried by the background shift rather than by a scale factor), into which the bubble is cut by interpolation:

$$\alpha = 1, \qquad \gamma_{ij} = \delta_{ij}, \qquad \beta^i = -\big(1 - f(r_s)\big)\,\frac{x^i}{L} - f(r_s)\,v^i, \qquad L = H^{-1}, \tag{57}$$

with f the same tanh top-hat used by the compact family and $\Lambda = 3H^2$. The bubble co-moves with the Hubble flow, $r(t) = r_0 e^{Ht}$ and $v = Hr(t)$, which is the paper's matching condition; substituting this relation into (57) gives $\beta = -x/L + f(r_s)\,(x - x_s)/L$, a sum of two radial gradients and hence *irrotational*. By Lemma 1 the Eulerian momentum density then vanishes identically and the wall is Hawking–Ellis Type I, which is the second case of Lemma 3. The wall fractions in Section 3.6 therefore use 100.0% Type I and 0.0% Type II. This geometry has a de Sitter background and is not asymptotically flat.

For the standardized row we choose $H = 0.02/15\,\mathrm{m}^{-1}$, $r_0 = 15\,\mathrm{m}$, $R_b = 15\,\mathrm{m}$, $\sigma = 0.497115373\,\mathrm{m}^{-1}$ and $t = 0$, so $v = Hr_0 = 0.02$. For the reference row we choose $H = 0.1\,\mathrm{m}^{-1}$, $r_0 = 1\,\mathrm{m}$, $R_b = 1\,\mathrm{m}$, $\sigma = 8\,\mathrm{m}^{-1}$ and $t = 0$, so $v = 0.1$. The relation $v(t) = Hr_0 e^{Ht}$ comes from Ref. [10]; these numerical parameter values are choices made here. We use $G = c = 1$, with time measured in meters and $\Lambda = 3H^2$.

E Deciding the conditions without an eigendecomposition

The energy conditions can be expressed as positivity conditions on the causal cone. In particular, the DEC requires $-T^a{}_b$ to preserve the future cone. Lorentz-cone-preserving maps and their semidefinite descriptions are established subjects [59, 60, 66]; Bergqvist and Senovilla [61] characterize tensors with the dominant property. Here we apply the classical S-lemma to obtain the four pointwise criteria, specifying the multiplier sign for each condition.

Proof of Theorem 1.

Proof. By (21) the WEC is exactly $q \geq 0$ on the open unit ball, hence by continuity on the closed ball, whose boundary is the NEC. Put $g(w) = 1 - |w|^2$. Since $g(0) = 1 > 0$, Slater's condition holds for the single quadratic constraint $g \geq 0$; and the S-lemma [64, 65] gives: $q \geq 0$ wherever $g \geq 0$ if and only if $q - \sigma g \geq 0$ on all of $\mathbb{R}^3$ for some $\sigma \geq 0$. Writing $q(w) - \sigma g(w) = (1, w)M(\sigma)(1, w)^{\mathsf{T}}$, global nonnegativity of that affine-argument quadratic is equivalent to $M(\sigma) \succeq 0$ (vectors with nonzero first component by scaling, the rest by continuity). This is (ii).

Part (i) is the equality-constrained case, and it is obtained by homogenizing. Put $x = (x_0, \tilde{w}) \in \mathbb{R}^4$ and

$$\tilde{q}(x) = \rho_n\, x_0^2 - 2x_0\, b{\cdot}\tilde{w} + \tilde{w}^{\mathsf{T}} S \tilde{w}, \qquad \tilde{g}(x) = x_0^2 - |\tilde{w}|^2,$$

homogeneous quadratic forms on $\mathbb{R}^4$ with $\tilde{q}(1, w) = q(w)$ and $\tilde{g}(1, w) = g(w)$. The cone $\{\tilde{g} = 0\}$ is $\{c\,(\pm 1, \hat{s}) : |\hat{s}| = 1,\ c \in \mathbb{R}\}$, and $\tilde{q}$ is even, so "$\tilde{q} \geq 0$ on $\{\tilde{g} = 0\}$" is exactly the null energy condition at the point. The equality-constrained S-lemma states that for homogeneous quadratic forms with $\tilde{g}$ indefinite [65, 68],

$$\tilde{q} \geq 0 \text{ on } \{\tilde{g} = 0\} \iff \exists\, \sigma \in \mathbb{R} :\ \tilde{q} - \sigma \tilde{g} \geq 0 \text{ on } \mathbb{R}^n,$$

the multiplier now free in sign because the constraint is an equality. Here $\tilde{g}$ is indefinite since $\tilde{g}(1, 0) = 1$ while $\tilde{g}(0, e_1) = -1$. The multiplier must be allowed either sign: for $\widehat{T} = \mathrm{diag}(-1, 2, 2, 2)$ the null contraction is $q(s) = 1 > 0$, so the null energy condition holds, yet $M(\sigma) \succeq 0$ only for $\sigma \in [-2, -1]$. Finally $\tilde{g}$ has matrix $-\eta$, so $\tilde{q} - \sigma\tilde{g}$ has matrix $\widehat{T} + \sigma\eta = M(\sigma)$ of (22), and nonnegativity of that form is $M(\sigma) \succeq 0$. This is (i).

For (iii), with the unnormalized $u^a = n^a + w^i e_i^a$ one has $g_{ab}u^a u^b = -1 + |w|^2$ and hence $\Theta_{ab}u^a u^b = q(w) + \frac{1}{2}T\, g(w)$, so the SEC is the WEC for Θ. For (iv), the DEC requires $T_{ab}u^a u^b \geq 0$ together with $J^a = -T^a{}_b u^b$ causal; and $J_a J^a = (T^2)_{ab}u^a u^b$, so causality of J is the WEC for $-(T^2)_{ab}$. Future-directedness needs no separate test: since $T_{ab}u^a u^b = -J_a u^a$, a causal J with the WEC satisfied is future-directed or zero. □

In a fixed tetrad, the optimized multiplier margin equals half the normalized null-cone minimum.

Corollary 2 (The multiplier margin is half the null-cone minimum)**.** *In a fixed orthonormal tetrad, writing $k^a = n^a + s^i e_i^a$ with $|s| = 1$,*

$$2 \max_{\sigma \in \mathbb{R}} \lambda_{\min}\big(M(\sigma)\big) \;=\; \min_{|s|=1} T_{ab}k^a k^b, \tag{58}$$

and the maximum is attained.

Proof. Write ν for the right-hand side. For any σ and any unit s the vector $x = (1, s)/\sqrt{2}$ has Euclidean norm one, and since $\tilde{g}(1, s) = 0$,

$$x^{\mathsf{T}} M(\sigma)\, x = \tfrac{1}{2}\big[\tilde{q}(1, s) - \sigma\tilde{g}(1, s)\big] = \tfrac{1}{2}\, T_{ab}k^a k^b,$$

so minimizing over s gives $\lambda_{\min}(M(\sigma)) \leq \nu/2$ for every σ. Conversely $\widehat{T} - \frac{\nu}{2}\mathbb{1}$ satisfies the null condition: a null $(1, s)$ has Euclidean square 2, so its contraction drops by exactly ν and is nonnegative by the definition of ν. Part (i) then supplies a σ with $\widehat{T} - \frac{\nu}{2}\mathbb{1} + \sigma\eta \succeq 0$, that is $\lambda_{\min}(M(\sigma)) \geq \nu/2$. Attainment follows from $\lambda_{\min}(M(\sigma)) \leq -|\sigma| + \lambda_{\max}(\widehat{T})$, which sends the concave function $\sigma \mapsto \lambda_{\min}(M(\sigma))$ to $-\infty$ at both ends. □

Multiplier search and exact certificates. M is affine in σ, so $\lambda_{\min}(M(\sigma))$ is *concave* and the search for a multiplier is a one-dimensional concave maximization. We use a ternary search, evaluating a 4×4 symmetric eigenvalue problem at each step. For rational input components, a rational σ together with an exact LDL^{T} factorization of $M(\sigma)$ is a certificate a reader can check by hand. This is a sufficient test. At a saturated optimum the admissible multiplier can be a single irrational value; an unsuccessful interval test remains inconclusive, and exact algebraic evaluation may be needed.

For example, with $\rho_n = 3$, $b = (1, 0, 0)$ and $S = \nVdash$,

$$\widehat{T} = \begin{pmatrix} 3 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad M(1) = \widehat{T} + \eta = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}, \tag{59}$$

and $M(1) = LDL^{\mathsf{T}}$ with L unit lower triangular carrying the single entry $L_{10} = -\frac{1}{2}$ and $D = \operatorname{diag}(2, \frac{3}{2}, 2, 2)$. Every entry is rational and every diagonal entry of D is positive, so $M(1) \succeq 0$ and the multiplier $\sigma = 1 \geq 0$ certifies the null and weak conditions at that point. The certificate can be checked by multiplying out LDL^{T}. For comparison, the admissible multipliers here are $\sigma \in [1 - \sqrt{3},\, 1 + \sqrt{3}]$, so the interval of rational choices is wide and $\sigma = 1$ is an interior point rather than a saturating one.

For the WEC, when no nonnegative multiplier exists, a violating observer can be sought by minimizing q on the ball directly. A negative-eigenvalue eigenvector of $M(\sigma_\star)$ need not give a violating observer. Writing that eigenvector as $(u_0, \mathbf{u})$ and taking $w = \mathbf{u}/u_0$ when $u_0 \neq 0$, one has $q(w) = \sigma(1 - |w|^2) + \lambda_{\min}|(u_0, \mathbf{u})|^2/u_0^2$, the first term being nonnegative for $\sigma \geq 0$ and $|w| \leq 1$, so the negative term has to outweigh it, which is not guaranteed by a negative eigenvalue alone. That construction also fails outright at a repeated $\lambda_{\min}$, where the eigensolver returns an arbitrary basis of the eigenspace: for $\widehat{T} = \operatorname{diag}(1, -3, 10, 10)$ the optimum is $\sigma_\star = 2$ with $M = \operatorname{diag}(-1, -1, 12, 12)$, and the routine can select e_0, returning $w = 0$ with $q(0) = +1$, while $w = (-0.999, 0, 0)$ gives $q \approx -1.994$. Minimizing q on the ball avoids both failures, and it does so without needing the minimum to be global: any w with $|w| \leq 1$ and $q(w) < 0$ refutes the condition: an interior point gives a timelike observer, and a negative boundary value persists at nearby interior points by continuity. The search is therefore one-sided by construction, and a failed witness search remains inconclusive; satisfaction requires a valid multiplier certificate. A quadratic on a ball can have a local minimizer that is not global: $q = -x^2 + x + y^2 + z^2$ has a strict local minimum at $w = +e_1$ with $q = 0$, against the global -2 at $w = -e_1$. A descent that stalled there would return no witness, so it must return an inconclusive result.

A proposed multiplier is checked in exact arithmetic. A proposed timelike violating vector can likewise be checked by direct contraction. The check can also use coordinate components. With e the matrix of tetrad vectors, $M(\sigma) = e(T + \sigma g)e^{\mathsf{T}}$ is congruent to the coordinate matrix, and congruence preserves positive semidefiniteness.

For the null condition, whose multiplier is free in sign, a matrix certificate can establish violation without explicitly finding a null vector. Write $\langle A, X\rangle = \operatorname{tr}(A^{\mathsf{T}}X)$ for the matrix pairing. To refute every σ at once a dual variable $X \succeq 0$ must satisfy $\langle g, X\rangle = 0$ *exactly*, since then $0 \leq \langle T + \sigma g, X\rangle = \langle T, X\rangle$ for every σ, so $\langle T, X\rangle < 0$ leaves no multiplier available. A rank-one $X = kk^{\mathsf{T}}$ would need a rational null k, and a rational Lorentzian form need not represent zero: over $\mathbb{Q}$ the form $\operatorname{diag}(-15, 1, 1, 1)$ has no nonzero null vector, because 15 is not a sum of three rational squares. Indeed, clearing denominators gives $a^2 + b^2 + c^2 = 15d^2$ in integers. After removing common factors of two, d must be odd: if d were even, reduction modulo four would force a, b, c even. For odd d, reduction modulo eight would require three squares to sum to seven, which is impossible. A rank-two X can instead provide a dual certificate. Take rational k, ℓ whose norms straddle the cone, $g(k,k)\, g(\ell,\ell) < 0$, and assume that the coordinate matrices g and T are rational. Set

$$X = \alpha\, kk^{\mathsf{T}} + \beta\, \ell\ell^{\mathsf{T}}, \qquad \alpha = |g(\ell,\ell)|, \quad \beta = |g(k,k)|. \tag{60}$$

Then $\langle g, X\rangle = \alpha\, g(k,k) + \beta\, g(\ell,\ell) = 0$ identically, by the opposite signs, while $\langle T, X\rangle = \alpha\, T(k,k) + \beta\, T(\ell,\ell)$ is a rational number whose negativity refutes every multiplier at once; a nonnegative value refutes nothing. Straddling alone is not sufficient: at least one contraction must be negative, and the weighted sum must be negative. For $\widehat{T} = \operatorname{diag}(1, -3, 0, 0)$ and $\ell = (0, 0, 1, 0)$, the timelike $k = (5, 4, 0, 0)$ gives $\langle T, X\rangle = -23$ and refutes the null condition, whereas $k = (2, 1, 0, 0)$ gives $+1$ and refutes nothing.

Pointwise and spatial bounds. Theorem 1 is *pointwise*: it asserts $\forall x\, \exists\, \sigma(x)$. It does not lift to a spatial box, where a single interval LMI would prove the strictly stronger $\exists\, \sigma\, \forall x$. That stronger statement already fails on the saturated family $\rho = a(x) > 0$, $b = 0$, $S = -a(x)\mathbb{1}$, for which $q \geq 0$ everywhere while $\sigma = a(x)$ is forced pointwise, so no common multiplier exists on any box over which a varies, and subdivision does not repair it. Global coverage over a domain is therefore left with the interval branch and bound of Appendix C, which brackets the sphere minimum $\min_{|w|=1} q$ directly.

Axial and full null minima. A single axial witness need not minimize the null contraction. For the canonical Type II block $(\mu, f, p_2, p_3) = (0, 1, -2, 0)$ it is zero, whereas $k = (1, 0, 1, 0)$ gives $T_{ab}k^a k^b = -1$. On the unit sphere the contraction is $3s_x^2 - 2s_x - 1 + 2s_z^2$, whose minimum $-4/3$ occurs at $s_x = 1/3$, $s_z = 0$. Theorem 1 gives the fixed-tetrad margin $-2/3$, as required by (58). The null-dust example has zero for both the axial witness and the null-cone minimum, so it does not distinguish the two diagnostics. Similarly, an imaginary-eigenvalue diagnostic supplies no quantitative margin for Type III, whose exact eigenvalues are real.

Numerical classification near degeneracies. The eigenvalue classifier depends on its tolerance near defective tensors. For the perturbed Jordan families used here, a perturbation δ displaces an eigenvalue by $\delta^{1/m}$ for block size m, so the dependence is not Lipschitz at $\delta = 0$. Fits over $\delta \in [10^{-14}, 10^{-8}]$ give slopes 0.509 for J_2 and 0.334 for J_3, close to 1/2 and 1/3. On the exact $J_3(-1) \oplus [3]$ test tensor, float64 gives a complex split of 5.7×10^{-6}; the classifier reports Type IV at tol $= 10^{-10}$, Type II at 2×10^{-6} and Type III at 5×10^{-6}. These are observations for the stated test tensor and solver. No fixed numerical precision resolves every exact degeneracy. At 50 digits the test tensors are correctly separated using the Jordan defect of the repeated eigenvalue.

We therefore report numerical type labels and detected disagreements as diagnostics. The energy-condition decision uses the LMI theorem and separately checked certificates. Garattini–Zatrimaylov uses its exact Type-I result. A fraction below 100% for WEC violations among numerically labeled Type III points would reveal an inconsistency between the numerical routes; it would not by itself identify which route failed.

Theorem 1 applies even when numerical classification is inconclusive; its numerical evaluation remains subject to the stated noise floor. For the Type III tensor (13), direct contraction gives an exact verdict: $T_{ab} = \ell_a m_b + m_a \ell_b$ gives $T_{ab}k^a k^b = 2(\ell\cdot k)(m\cdot k)$, so the rational null direction $k^a = (1, 0, 1, 0)$ returns -2, and no multiplier can certify a condition that a rational null vector violates.

The type also supplies a consistency check. Types III and IV violate *every* standard energy condition [5, 14]: for a Type III tensor the null condition cannot be satisfied once the off-diagonal Jordan entry is nonzero, so a nontrivial Type III point violates all four. Any point the classifier labels III or IV at which the LMI numerically reports satisfaction gives a numerically detected label contradiction. In the separate regularized WarpShell diagnostic at $v_s = 0.5$, the full 50^3 grid over $[-25, 25]^3$ contains 92 616 Type I and 32 384 Type IV points and no Type II or Type III point at all. The WEC diagnostic is negative at all 32 384 Type-IV points, and no NEC margin exceeds the positive noise floor; the check therefore records zero detected contradictions. That verdict is reported and not certified: the multiplier search runs in binary64 and $\lambda_{\min}$ from a symmetric eigensolver carries a backward error of order $\varepsilon\|M\|$, so a margin is a verdict only once it clears a noise floor in either direction. That floor is

$$\epsilon_{\text{floor}} = 10^{-12} \max_{IJ} |\widehat{T}_{IJ}| + 10^{-18}, \tag{61}$$

relative to the scale of the tensor, with a small absolute term that keeps exact vacuum, where the residual ternary bracket returns about -6×10^{-22}, from being reported as a violation. Inside the floor the answer is *inconclusive*, and a separate exact check may resolve it: a rational multiplier checked by an exact LDL^{T}, or a checked violating-vector or dual certificate, as appropriate to the condition.

The same check over the reported family. The same test applied to the four constructions of this paper, at every point of the wall-clustered 48^3 benchmark grid and at $v_s = 0.5, 1, 2$, is Table 16(a). The eigenvalue and LMI routes agree on 100.00% of decisive Type-I comparisons in every row. The exact criteria are equivalent, so a numerical disagreement could arise in either implementation. The fraction of points labeled Type III or IV at which the inequality numerically reports *satisfaction*, which by the argument above cannot be reconciled with the label, is zero over

Table 16. *(a)* Type-free check of the type-based route on the wall-clustered 48^3 benchmark grid. *Left*: agreement of Theorem 1 with the eigenvalue inequalities among decisive Type-I comparisons. *Right*: Type-III and Type-IV points at which the inequality numerically reports satisfaction. Both thresholded against (61); Rodal has no non-Type-I point, so its right-hand entries are undefined. *(b)* Crossing the Type-I/Type-IV transition on the analytic families of the text. "Labeled II" counts sweep points the float64 classifier returns as Type II. The maximum adjacent-grid slope is an observed statistic. The last column compares twice the numerical LMI margin with the minimum contraction over 400 000 sampled null directions; the sampling error is not enclosed.

		Type-I verdicts agreeing (%)			Numerically detected label contradictions (%)		
	Metric	$v_s = 0.5$	1.0	2.0	$v_s = 0.5$	1.0	2.0
(a)	Alcubierre	100.00	100.00	100.00	0.00	0.00	0.00
	Natário	100.00	100.00	100.00	0.00	0.00	0.00
	Van den Broeck	100.00	100.00	100.00	0.00	0.00	0.00
	Rodal	100.00	100.00	100.00	–	–	–

	Family	points	labeled II	max. grid slope	$\lvert 2\mathcal{N}_{\mathrm{LMI}} - \min_{s\in\mathrm{sample}} q(s)\rvert$
(b)	Momentum, $\Delta \to 0$	401	1	1	2.8e-06
	Momentum, decoupled	401	1	0.499	2.0e-09
	Transverse ($\Delta > 0$)	301	0	0.499	8.5e-05

619 288 such points across the family. Both are thresholded against (61) and not against zero. This does not make the classifier exact: the decisive comparisons on these sampled geometries show no disagreement, and the inequality decides every wall point at which the type-based route returns no verdict (1 304 of 1 304 for Alcubierre at $v_s = 0.5$, and likewise at every other entry).

The separate WarpShell comparison has three limits. First, 89 808 of the 92 616 Type-I labels are near-vacuum points assigned Type I by the near-vacuum test rather than computed from an eigenstructure, so its discriminating power lies in the 2 808 resolved Type-I points and the 32 384 Type-IV ones. Second, the Type-III case tested *zero* points, so it supplies no evidence for that case. Third, the regularized WarpShell lies outside the matched benchmark family. A finite spatial grid need not sample a type-transition locus. The analytic tensor families below provide known examples on and across those loci.

Transitions in analytic tensor families. On analytic one-parameter families the algebraic type is known in closed form, an independent brute-force scan over 400 000 null directions is available at every point, and the transition can be crossed deliberately rather than sampled. Three families are swept (Table 16(b)), all in a Minkowski orthonormal frame so that the classifier, the inequality and the scan operate on the same matrix.

The first is the momentum block itself, $\rho = S_{\parallel} = 1$ with $p_\perp = 0$ and $|j|$ swept through 1, so that $\Delta = 4(1 - |j|^2)$ carries the point from Type I through the nilpotent $J_2(0)$ at $|j| = 1$ to Type IV. Across 401 samples the maximum observed adjacent-grid slope of the margin is 1. The analytic margin passes continuously through the locus ($+4.99 \times 10^{-3}$ at $|j| = 0.995$, exactly 0 at $|j| = 1$, -5.00×10^{-3} at $|j| = 1.005$), while the label is discontinuous there. Twice the margin differs from the finite-sample null minimum by at most 2.8×10^{-6}, an observed discrepancy from the angular scan.

The second family fixes the same Δ and the same Type II point but sets $p_\perp = -3$, which puts the null-cone minimum at $-2 - |j|^2/4$. The label still runs $\mathrm{I} \to \mathrm{II} \to \mathrm{IV}$ across the sweep while the sign of the margin does not change at any of the 401 points, so the algebraic type is not a proxy for an energy-condition verdict: the type changes while the NEC remains violated.

The third is the transverse family of Corollary 1, where $\Delta = 3 > 0$ throughout and Type IV is opened by the transverse coupling instead. The measured band endpoints, $m = 0.78$ and $m = 2.01$ at a sweep spacing of 0.01, bracket the predicted $m_\star \approx 0.782$ and $m_2 \approx 2.015$ within one sweep step.

A separate fourth family tests the Type III case. On an exact Segre $[3, 1]$ family, 121 points log-spaced over $|f| \in [10^{-6}, 1]$, the 50-digit recomputation returns Type III at every one. The float64 classifier returns Type IV at every one of the 121 at $\mathrm{tol} = 10^{-10}$; at $\mathrm{tol} = 2 \times 10^{-6}$ it returns Type II at 63, Type III at 40 and Type IV at 18, and at 5×10^{-6} it returns 70, 45 and 6. The label is a function of the tolerance across the whole family, and it is never right at the tolerance used here. The binary64 linear matrix inequality detects null-energy violation at all 121 tested points. The verdict therefore survives six decades of chain strength over which the label does not.

Table 17. The null-energy verdict taken back to the metric, at $R_b = 1$, $\sigma = 8$, $v_s = \frac{1}{2}$. $\mathcal{N}$ is the type-free null deficit (43). The interval curvature chain encloses the tensor; the lower endpoint follows from the interval LDL^{T} test and the upper endpoint from an interval-normalized null contraction. The two endpoints agree to the six decimals printed, so the width is given as its own column.

Metric	wall point	$\mathcal{N}_{\text{lower}}$	$\mathcal{N}_{\text{upper}}$	width	verdict
Alcubierre	on axis	-0.198944	-0.198944	7.7×10^{-15}	violated
Alcubierre	wall, $s = 0.35$	-0.343691	-0.343691	2.8×10^{-16}	violated
Alcubierre	wall, $s = 0.5$	-0.146460	-0.146460	1.1×10^{-16}	violated
Natário	on axis	-0.557042	-0.557042	3.3×10^{-16}	violated
Natário	wall, $s = 0.35$	-1.275487	-1.275487	8.9×10^{-16}	violated
Natário	wall, $s = 0.5$	-1.207662	-1.207662	6.7×10^{-16}	violated
Van den Broeck	on axis	$+0.000195$	$+0.000195$	4.5×10^{-15}	satisfied
Van den Broeck	wall, $s = 0.35$	$+0.034675$	$+0.034675$	1.3×10^{-16}	satisfied
Van den Broeck	wall, $s = 0.5$	-1.006795	-1.006795	4.4×10^{-16}	violated
Rodal	on axis	-0.156336	-0.156336	1.1×10^{-14}	violated
Rodal	wall, $s = 0.35$	-0.169108	-0.169108	1.7×10^{-16}	violated
Rodal	wall, $s = 0.5$	-0.000911	-0.000911	3.3×10^{-14}	violated

E.1 Rigorous pointwise bounds from the metric

An exact certificate applies to its input tensor. To establish an energy-condition verdict for a prescribed metric, the numerical error in that tensor must also be enclosed. Appendix C provides this step by interval evaluation of the curvature.

At a point, the interval curvature chain encloses the induced tensor. The pointwise tests below use the interval LDL^{T} lower bound: a trial spectral bound ℓ is accepted when the factorization establishes $M(\sigma) - \ell\mathbb{1} \succeq 0$. Then 2ℓ bounds $\mathcal{N}$ from below. An interval-normalized null contraction supplies the upper bound. Together these bounds determine the NEC sign whenever the resulting interval excludes zero.

Table 17 applies these bounds at three wall points of each drive at $v_s = \frac{1}{2}$: $(x, s) = (1, 0), (0.9, 0.35), (0.95, 0.5)$ in units of R_b. The brackets close to a width between 1.1×10^{-16} and 3.3×10^{-14} in absolute terms, which is ten significant digits on Van den Broeck's small on-axis margin and fourteen on the deep wall values, with the input tensor and spectral bounds enclosed by directed rounding. A floating-point eigensolver proposes σ; the interval LDL^{T} test establishes the bound for that proposal. A poor proposal can widen the bracket but cannot invalidate it. For Van den Broeck the NEC is certified satisfied at $(x, s) = (1, 0)$ and $(0.9, 0.35)$, and violated at $(0.95, 0.5)$.

The composition covers the null, weak, strong and dominant conditions alike: Θ_{ab} and $-(T^2)_{ab}$ are interval-algebraic rearrangements of the components (ρ_n, b_i, S_{ij}) returned by the chain, and by Theorem 1 the strong and dominant cases are the ball condition on those two tensors. For a null k, $\Theta(k, k) = T(k, k)$, so a strict null violation also implies SEC violation by continuity toward timelike vectors. The numerical SEC certificate evaluates a timelike witness for Θ directly. The weak-condition ball includes $w = 0$, so $\rho_n < 0$ gives a weak-condition witness without a direction search.

The strict interval test remains *inconclusive* at saturation. For exact vacuum, $q \equiv 0$ establishes satisfaction analytically.

E.2 Interval tests on a wall grid

Table 18 applies the interval tests to 4257 wall points per drive, on a 33×129 product grid in (r, θ) over $f \in [0.1, 0.9]$. All four conditions are decided at each sampled point.

The tests require neither a Hawking–Ellis classification nor a rapidity cap. They also apply on the Type II locus of (12), where numerical classification can be ill-conditioned. Across 17,028 points and 68,112 verdicts, no interval evaluation failed and no verdict remained inconclusive.

These counts describe a finite sample, with band membership certified at each point; they do not measure wall volume. The separate branch-and-bound brackets in Table 15 bound the NEC infimum over the whole wall.

The census and global search supply upper bounds on the same infimum from different points. Either upper bound can be smaller. For every drive, the census upper endpoint lies above the global lower endpoint, as required. Relative to the magnitude of the search upper endpoint, the two upper bounds differ by 0.17%, 0.084%, 1.14% and 0.020% for Alcubierre, Natário, Van den Broeck and Rodal, respectively. These observed differences do not bound the error in the global infimum.

Table 18. Every sampled wall point decided from the metric, at $R_b = 1$, $\sigma = 8$, $v_s = \frac{1}{2}$ (Van den Broeck also $\tilde{R} = 1$, $\alpha = \frac{1}{2}$, $\sigma_B = 8$), 80-bit working precision. Per construction, 4257 points of a 33×129 grid in (r, θ) over the band $f \in [0.1, 0.9]$ ($r \in [0.8627, 1.1373]$). Entries count *sampled points*, not wall measure. "Inconclusive" counts unresolved verdicts, which are not evidence that the condition holds. "Deepest upper" is the most negative certified upper endpoint attained.

Construction	condition	certified violated	certified satisfied	inconclusive	deepest upper
Alcubierre	NEC	4257	0	0	−0.6300
	WEC	4257	0	0	−0.6300
	SEC	4257	0	0	−0.6300
	DEC	4257	0	0	−0.6300
Natário	NEC	4257	0	0	−5.2169
	WEC	4257	0	0	−5.2169
	SEC	4257	0	0	−5.2169
	DEC	4257	0	0	−7.1228
Van den Broeck	NEC	3359	898	0	−1.2962
	WEC	3359	898	0	−1.2962
	SEC	4257	0	0	−1.2962
	DEC	3359	898	0	−1.2962
Rodal	NEC	3811	446	0	−0.1942
	WEC	3811	446	0	−0.1942
	SEC	3829	428	0	−0.2235
	DEC	4205	52	0	−0.1942

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